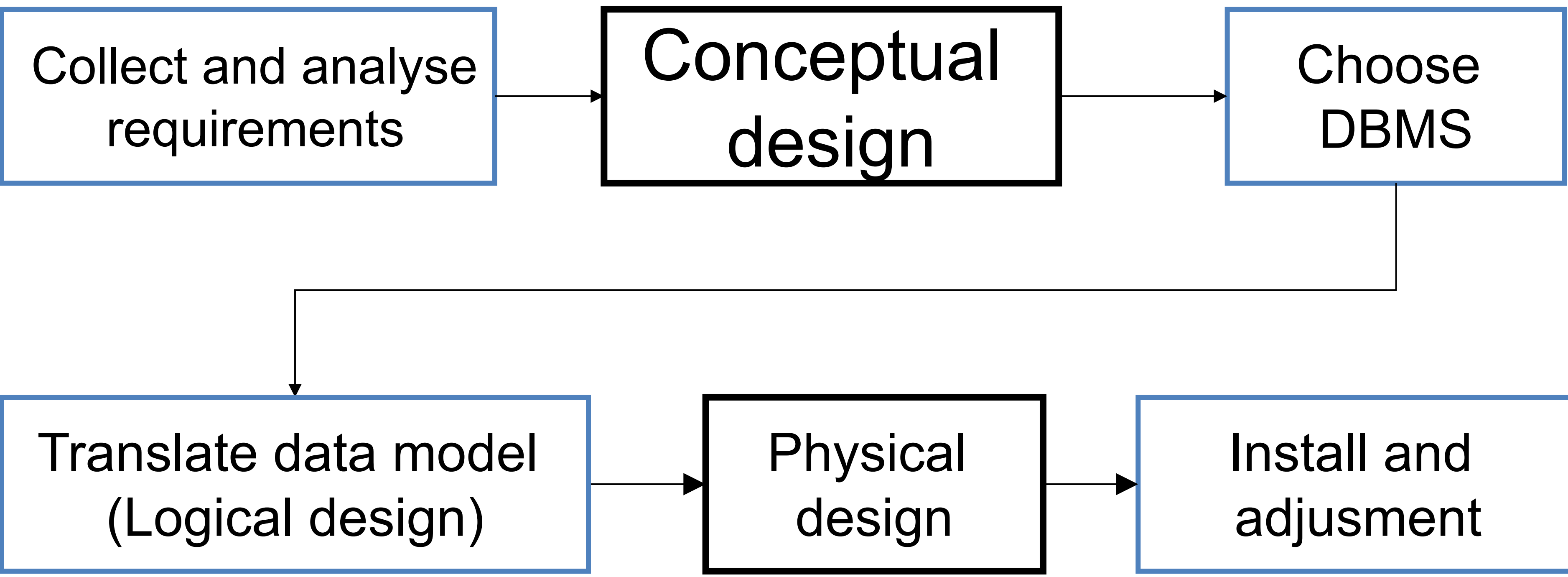
Entity-Relationship Model

E/R Diagrams

Weak Entity Sets

Converting E/R Diagrams to Relations

Database designing process



Purpose of E/R Model

- The E/R model allows us to sketch database schema designs.
  - Includes some constraints, but not operations.
- Designs are pictures called *entity-relationship diagrams*.
- Later: convert E/R designs to relational DB designs.

Framework for E/R

- Design is a serious business.
- The "boss" knows they want a database, but they don't know what they want in it.
- Sketching the key components is an efficient way to develop a working database.

Entity Sets

- *Entity* = “thing” or object.
- *Entity set* = collection of similar entities.
  - Similar to a class in object-oriented languages.
- *Attribute* = property of (the entities of) an entity set.
  - Attributes are simple values, e.g. integers or character strings, not structs, sets, etc.

Types of Attributes

- **Simple attribute** − are atomic values, which cannot be divided further. For example, a student's phone number is an atomic value of 10 digits.
- **Composite attribute** − are made of more than one simple attribute. For example, a student's complete name may have first_name and last_name.
- **Derived attribute** − are the attributes that do not exist in the physical database, but their values are derived from other attributes present in the database. For example, average_salary in a department should not be saved directly in the database, instead it can be derived. For another example, age can be derived from data_of_birth.

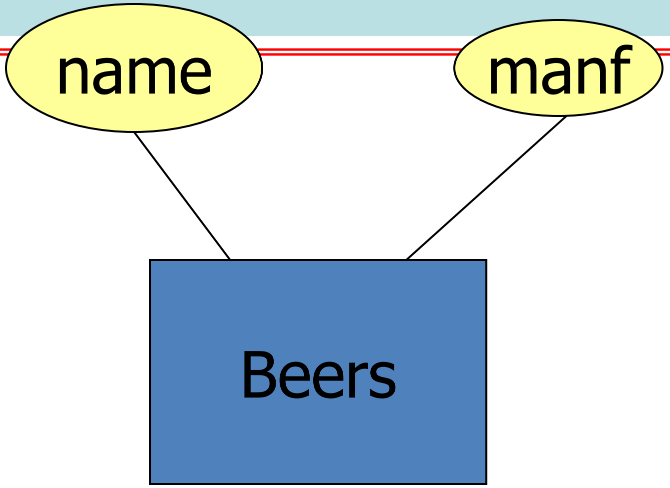
Types of Attributes

- **Single-value attribute** − Single-value attributes contain single value. For example − Social_Security_Number.
- **Multi-value attribute** − Multi-value attributes may contain more than one values. For example, a person can have more than one phone number, email_address, etc.

E/R Diagrams

- In an entity-relationship diagram:
  - Entity set = rectangle.
  - Attribute = oval, with a line to the rectangle representing its entity set.

Example:

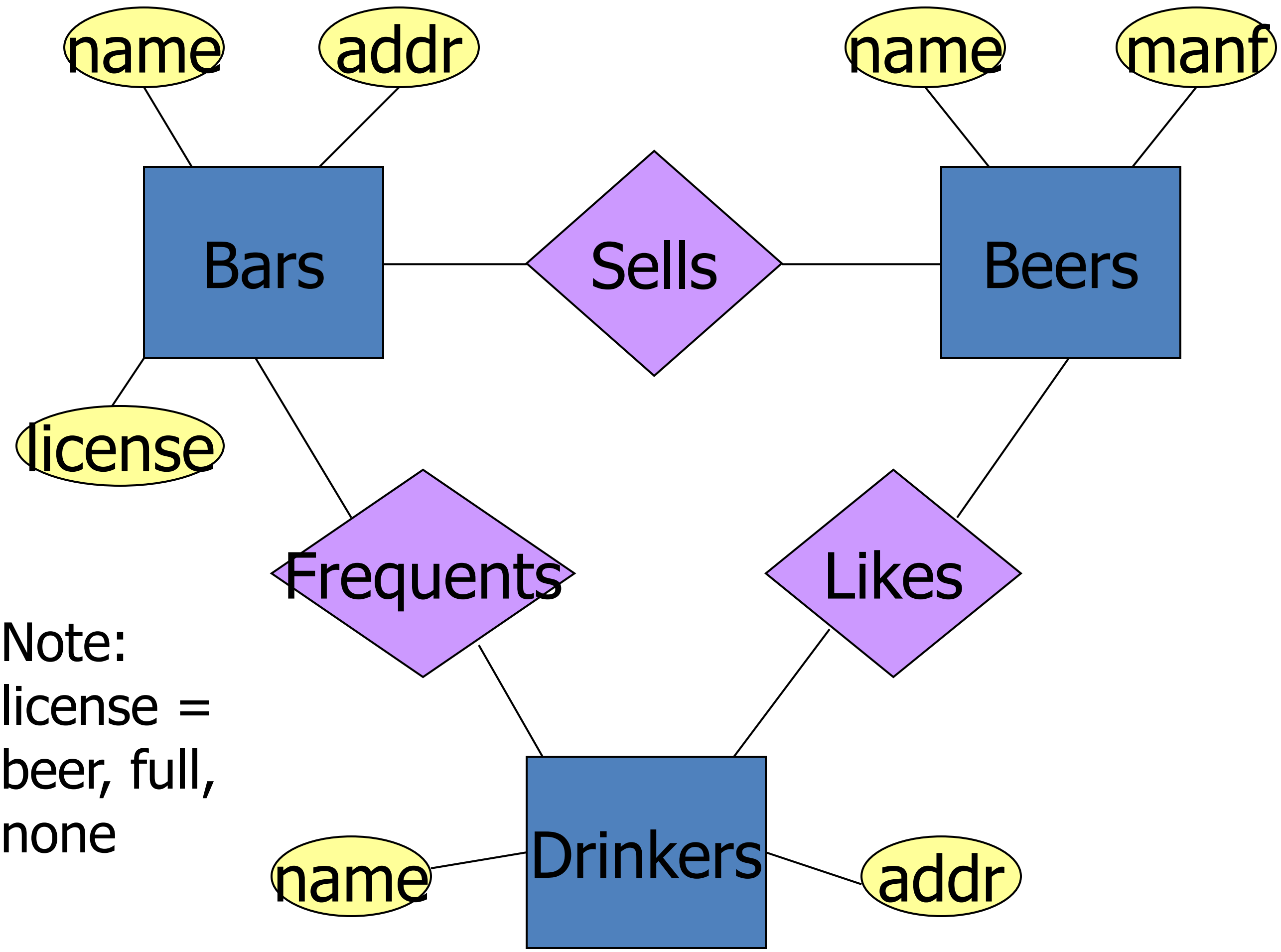


- Entity set Beers has two attributes, name and manf (manufacturer).
- Each Beers entity has values for these two attributes, e.g. (Bud, Anheuser-Busch)

Relationships

- A relationship connects two or more entity sets.
- It is represented by a diamond, with lines to each of the entity sets involved.

Example: Relationships



Note:
license =
beer, full,
none

Bars sell some beers.

Drinkers like some beers.

Drinkers frequent some bars.

Relationship Set

- The current “value” of an entity set is the set of entities that belong to it.
  – Example: the set of all bars in our database.
- The “value” of a relationship is a *relationship set*, a set of tuples with one component for each related entity set.

Example: Relationship Set

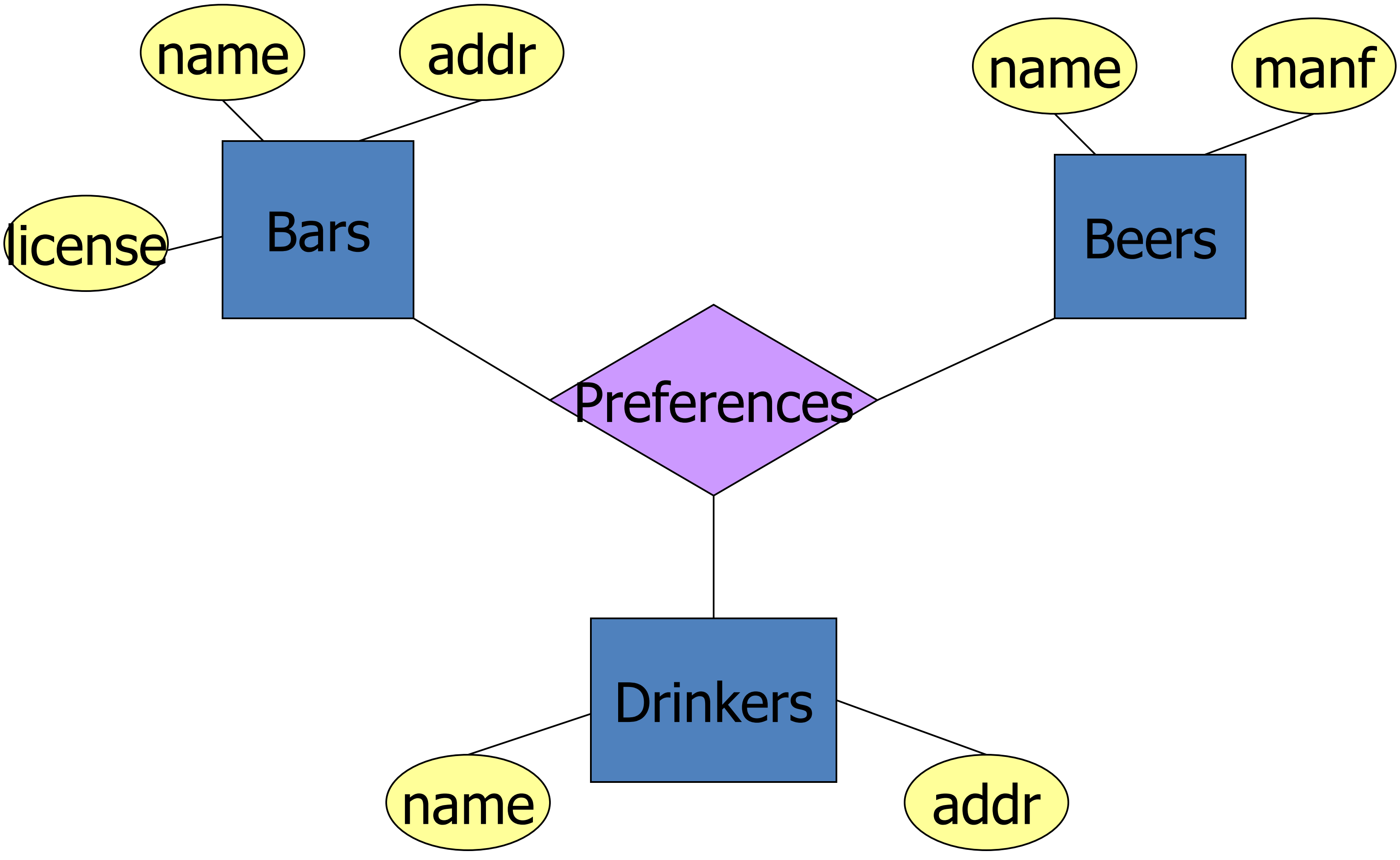
- For the relationship Sells, we might have a relationship set like:

Bar	Beer
Joe's Bar	Bud
Joe's Bar	Miller
Sue's Bar	Bud
Sue's Bar	Pete's Ale
Sue's Bar	Bud Lite

Multiway Relationships

- Sometimes, we need a relationship that connects more than two entity sets.
- Suppose that drinkers will only drink certain beers at certain bars.
  - Our three binary relationships Likes, Sells, and Frequents do not allow us to make this distinction.
  - But a 3-way relationship would.

Example: 3-Way Relationship



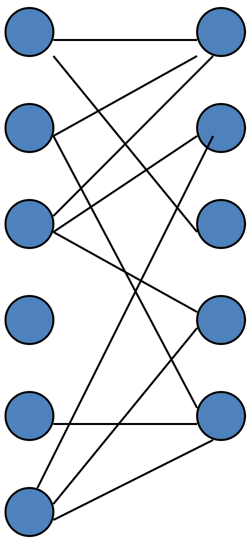
A Typical Relationship Set

Bar	Drinker	Beer
Joe’s Bar	Ann	Miller
Sue’s Bar	Ann	Bud
Sue’s Bar	Ann	Pete’s Ale
Joe’s Bar	Bob	Bud
Joe’s Bar	Bob	Miller
Joe’s Bar	Cal	Miller
Sue’s Bar	Cal	Bud Lite

Many-Many Relationships

- Focus: binary relationships, such as Sells between Bars and Beers.
- In a *many-many* relationship, an entity of either set can be connected to many entities of the other set.
  - E.g., a bar sells many beers; a beer is sold by many bars.

In Pictures:

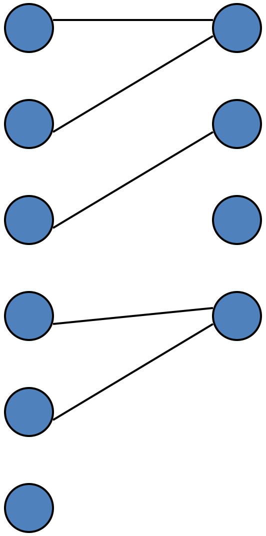


many-many

Many-One Relationships

- Some binary relationships are *many -one* from one entity set to another.
- Each entity of the first set is connected to at most one entity of the second set.
- But an entity of the second set can be connected to zero, one, or many entities of the first set.

In Pictures:



many-one

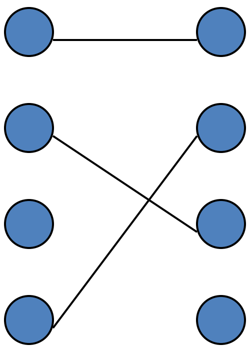
Example: Many-One Relationship

- Favorite, from Drinkers to Beers is many-one.
- A drinker has at most one favorite beer.
- But a beer can be the favorite of any number of drinkers, including zero.

One-One Relationships

- In a *one-one* relationship, each entity of either entity set is related to at most one entity of the other set.
- Example: Relationship Best-seller between entity sets Manfs (manufacturer) and Beers.
  - A beer cannot be made by more than one manufacturer, and no manufacturer can have more than one best-seller (assume no ties).

In Pictures:

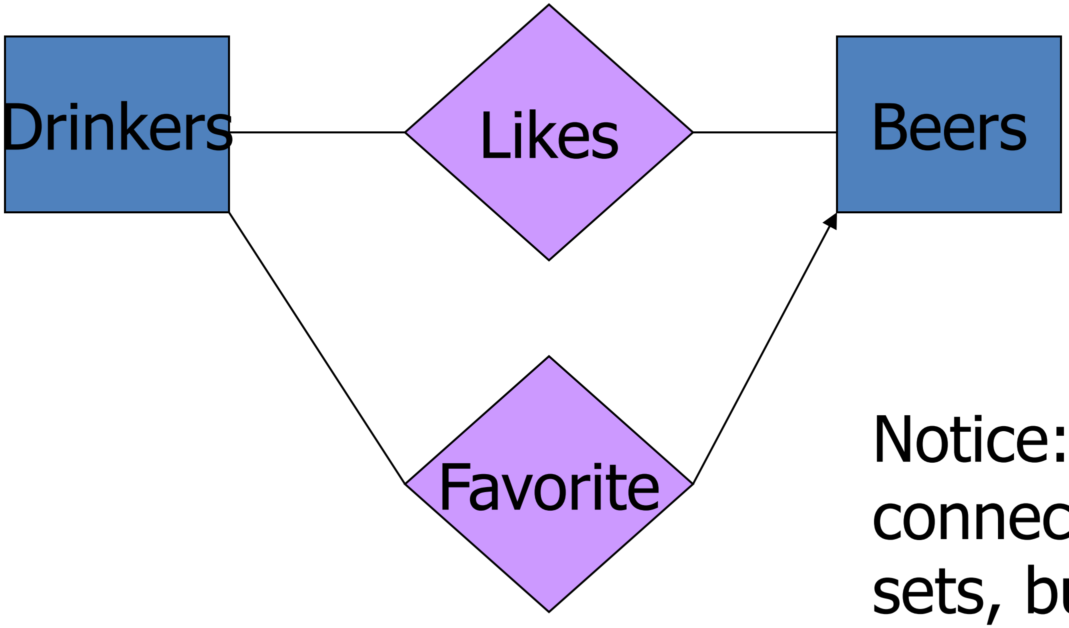


one-one

Representing “Multiplicity”

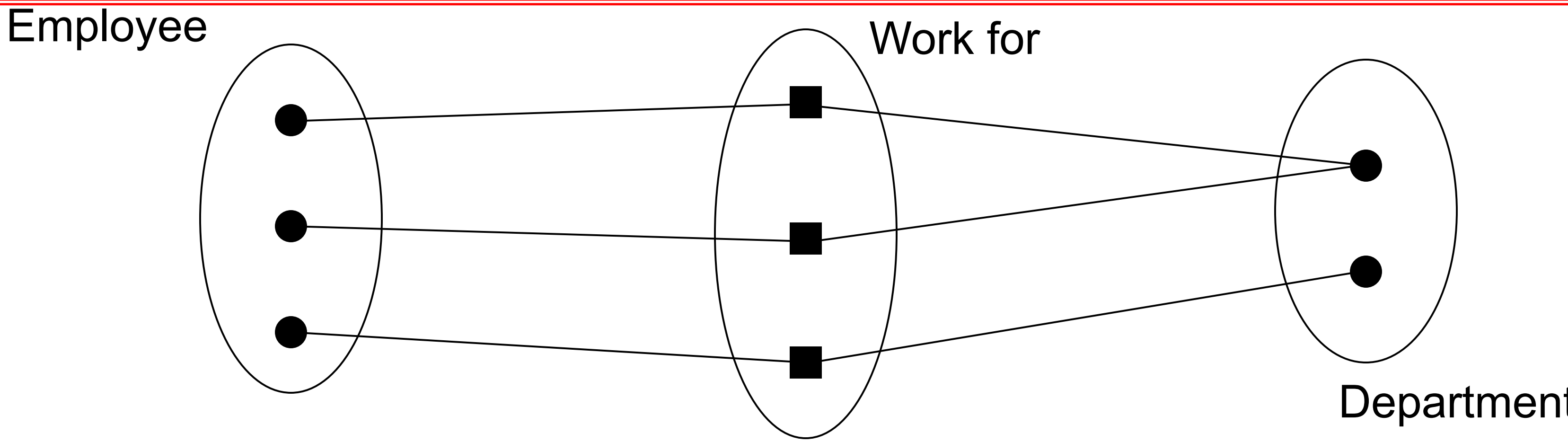
- Show a many-one relationship by an arrow entering the “one” side.
  - Remember: Like a functional dependency.
- Show a one-one relationship by arrows entering both entity sets.
- Rounded arrow = “exactly one,” i.e., each entity of the first set is related to exactly one entity of the target set.

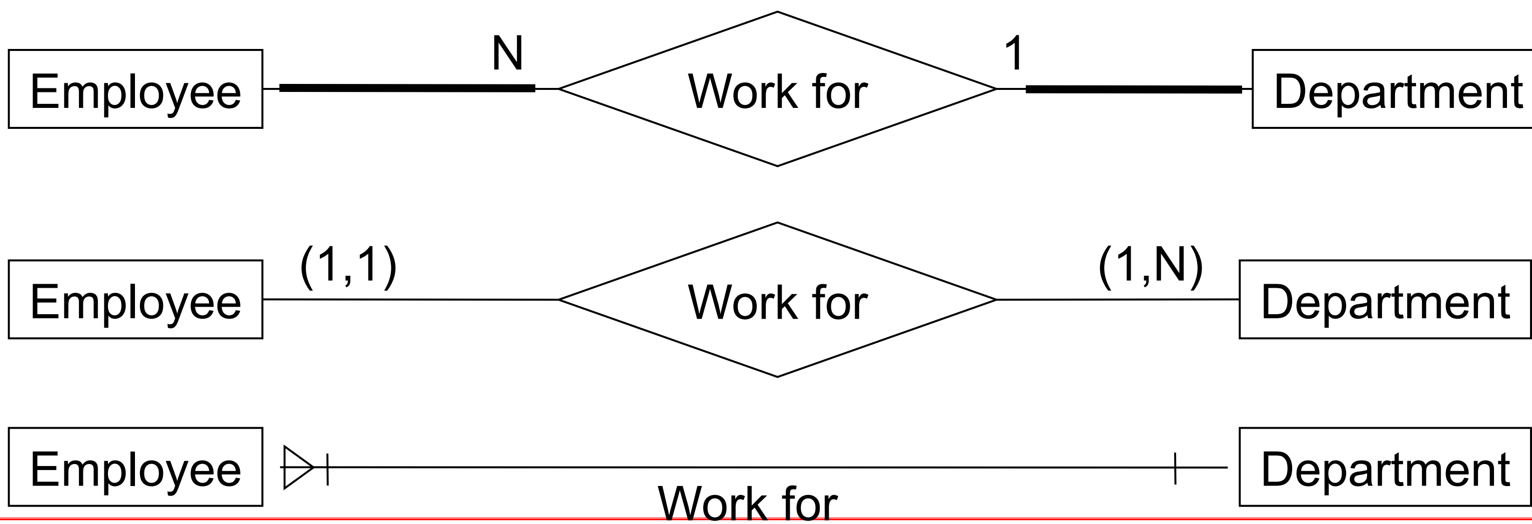
Example: Many-One Relationship



Notice: two relationships connect the same entity sets, but are different.

Many-One Relationship (1 : N)



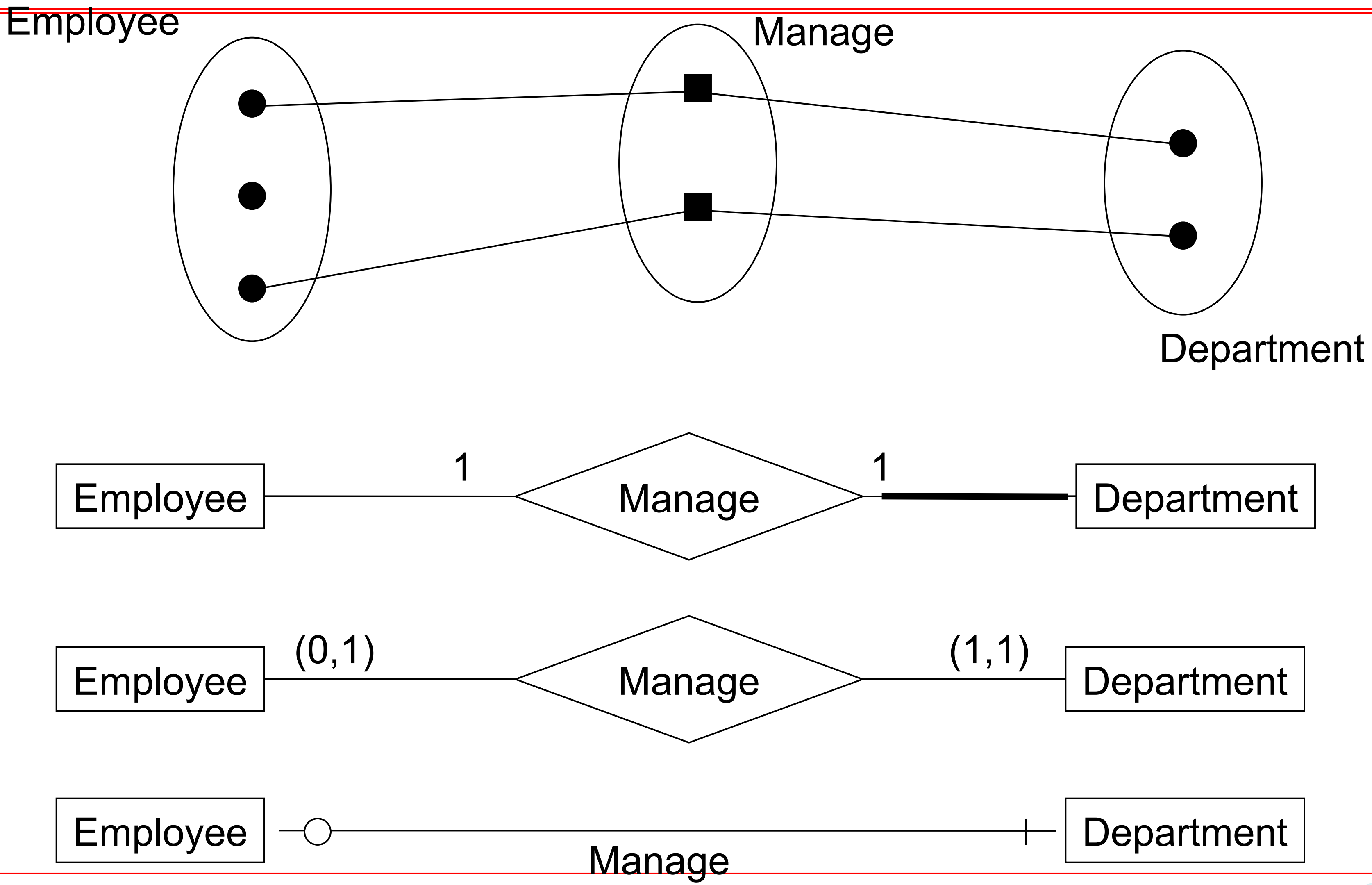


In the E/R Diagram

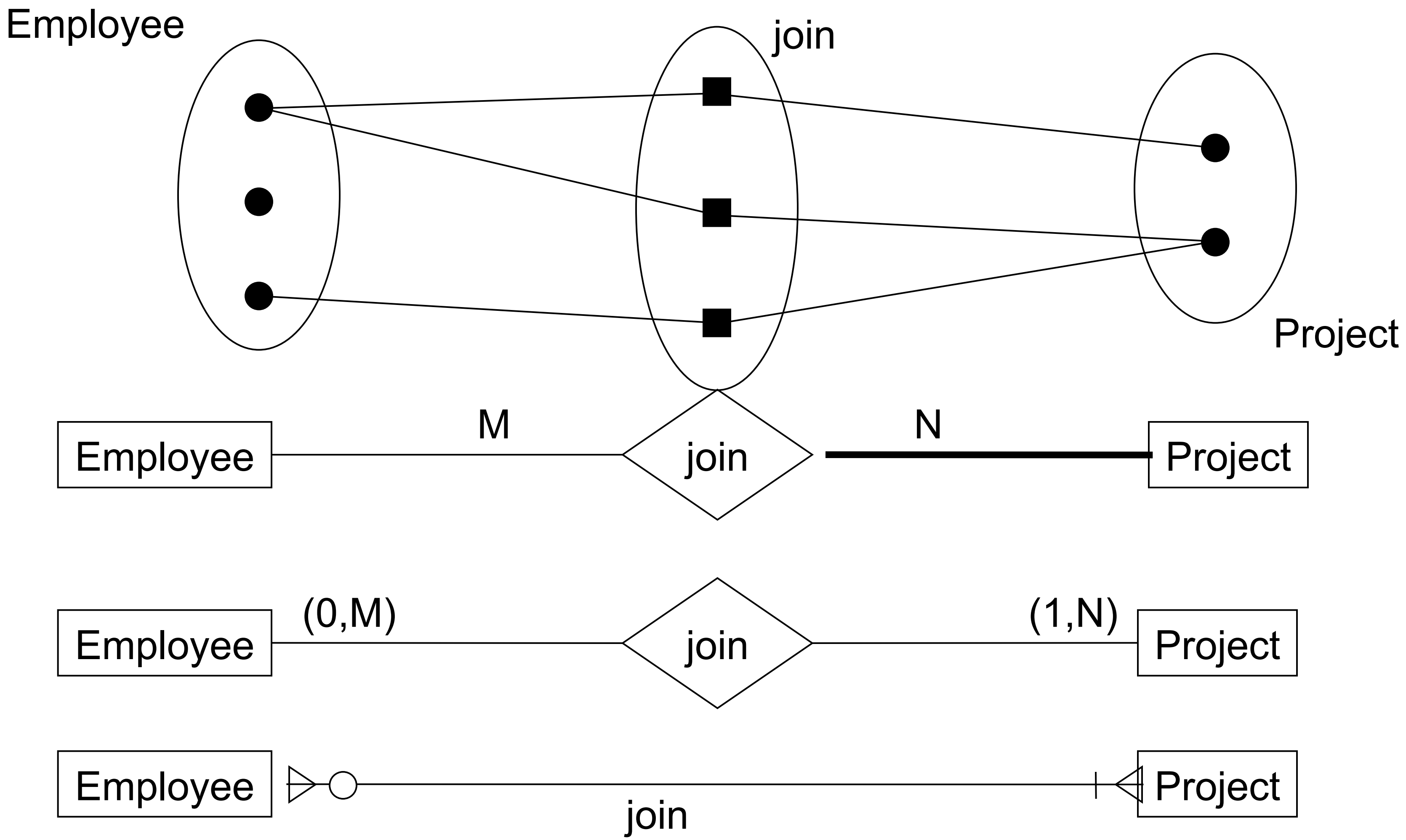
A beer is the best-seller for 0 or 1 manufacturer.

A manufacturer has exactly one best seller.

One-one relationship (1 : 1)

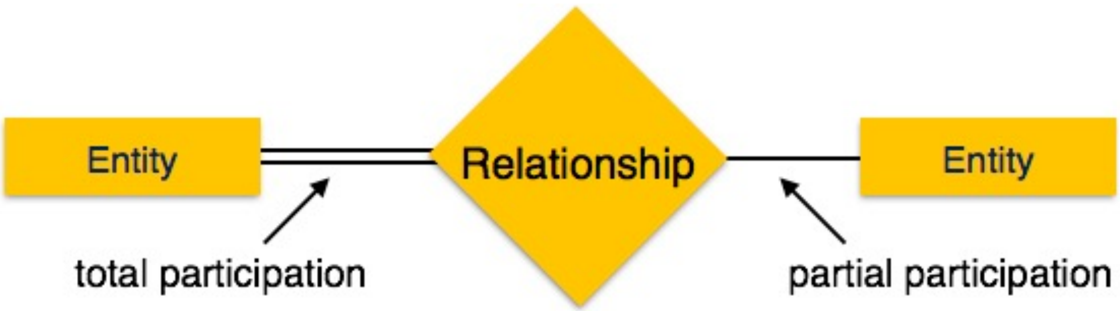


Many – many relationship (M : N)



Participation Constraints

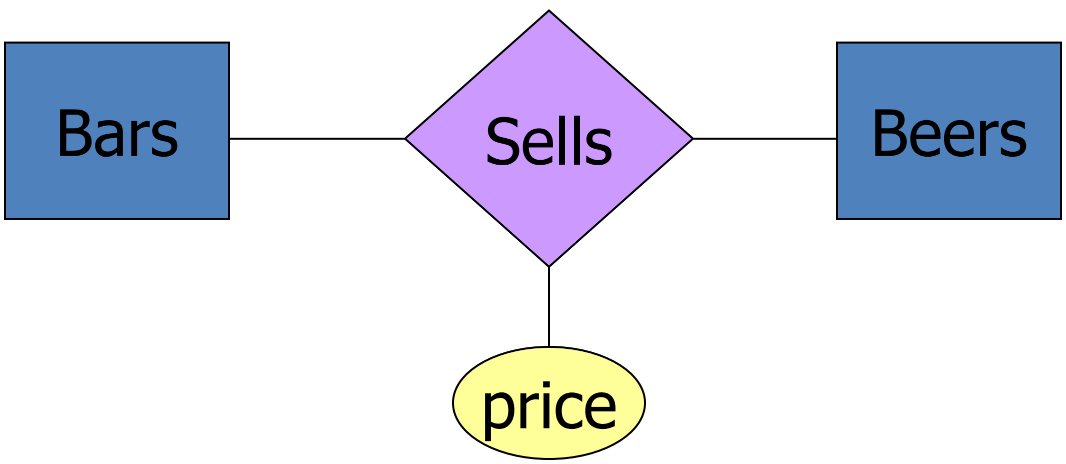
- **Total Participation** – Each entity is involved in the relationship. Total participation is represented by double lines.
- **Partial participation** – Not all entities are involved in the relationship. Partial participation is represented by single lines.



Attributes on Relationships

- Sometimes it is useful to attach an attribute to a relationship.
- Think of this attribute as a property of tuples in the relationship set.

Example: Attribute on Relationship

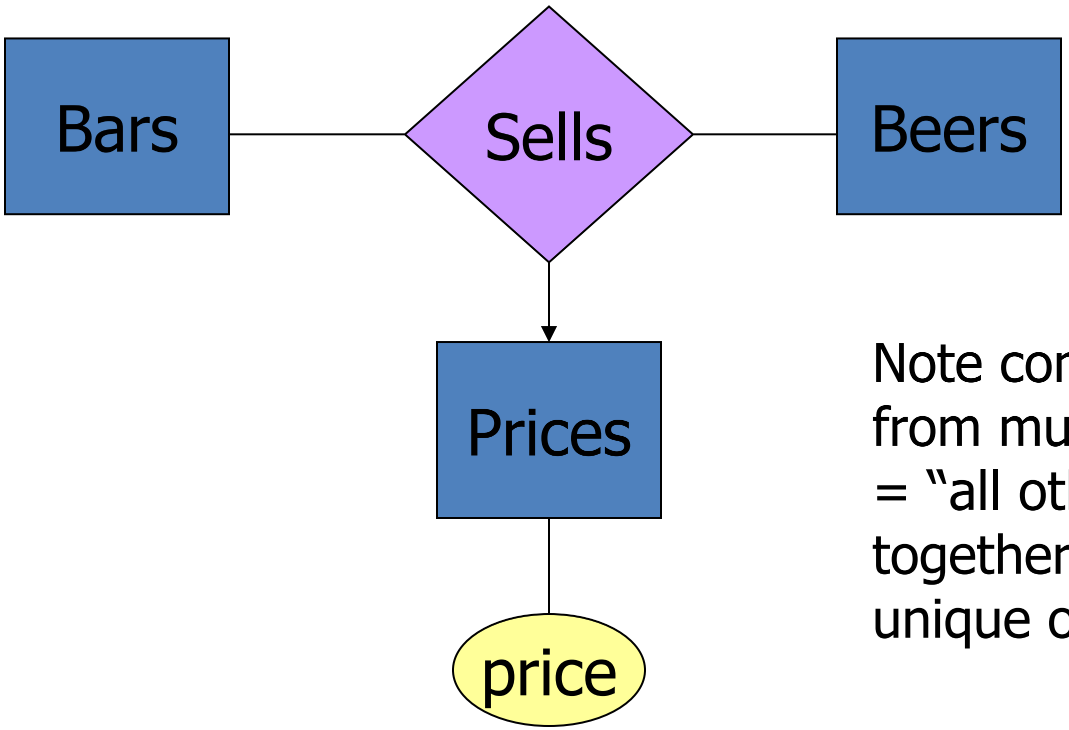


Price is a function of both the bar and the beer, not of one alone.

Equivalent Diagrams Without Attributes on Relationships

- Create an entity set representing values of the attribute.
- Make that entity set participate in the relationship.

Example: Removing an Attribute from a Relationship

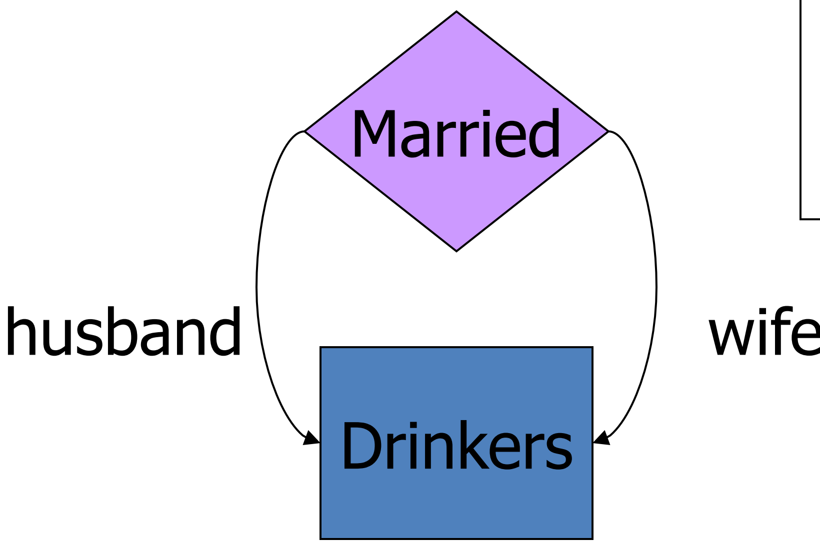


Note convention: arrow from multiway relationship = "all other entity sets together determine a unique one of these."

Roles

- Sometimes an entity set appears more than once in a relationship.
- Label the edges between the relationship and the entity set with names called *roles*.

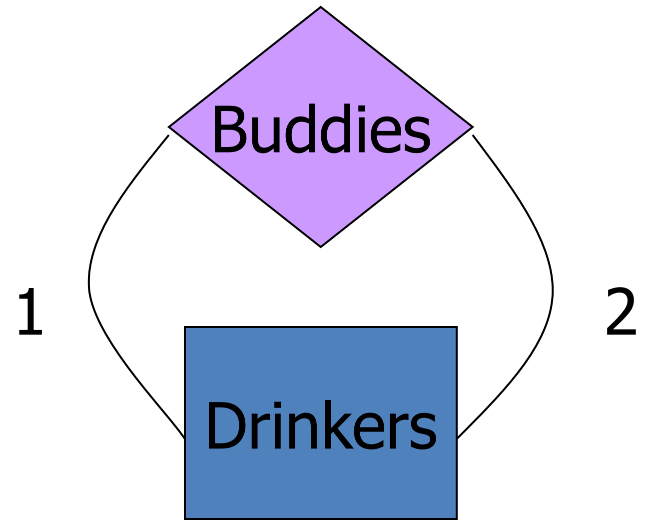
Example: Roles



Relationship Set

Husband	Wife
Bob	Ann
Joe	Sue
…	…

Example: Roles



Relationship Set

Buddy1	Buddy2
Bob	Ann
Joe	Sue
Ann	Bob
Joe	Moe
…	…

Subclasses

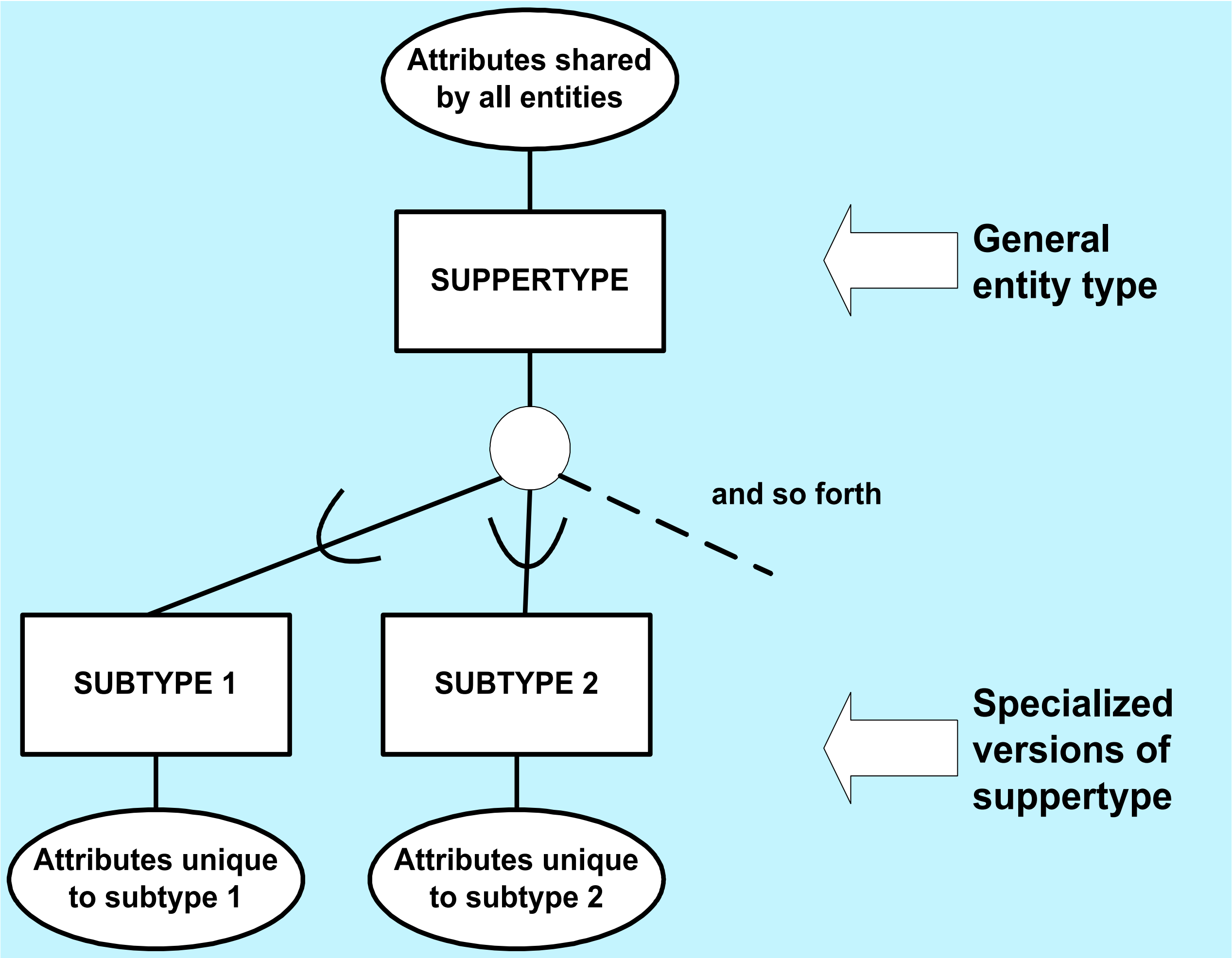
- *Subclass* = special case = fewer entities = more properties.
- Example: Ales are a kind of beer.
  - Not every beer is an ale, but some are.
  - Let us suppose that in addition to all the *properties* (attributes and relationships) of beers, ales also have the attribute color.

Subclasses in E/R Diagrams

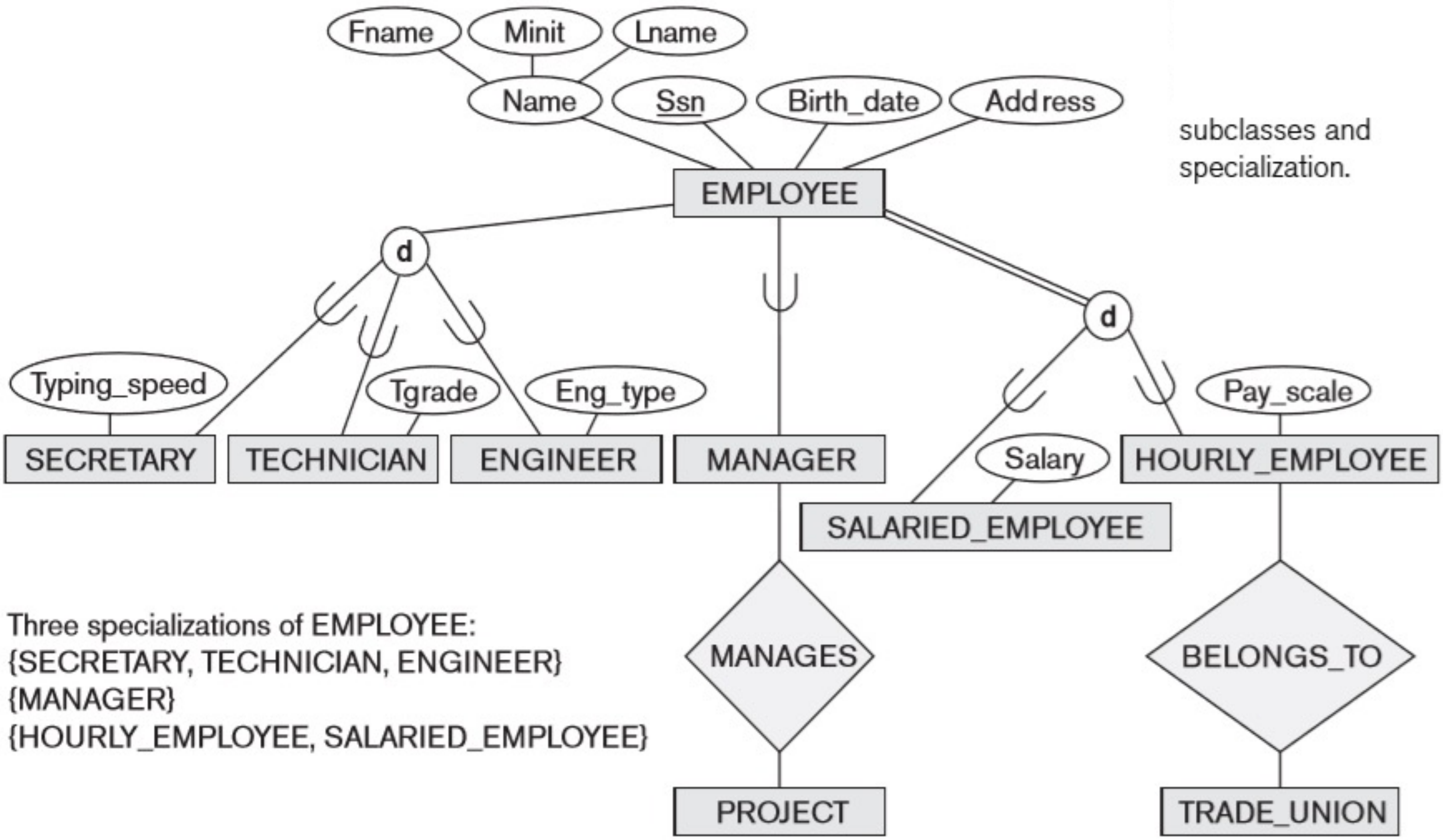
- Assume subclasses form a tree.
  - I.e., no multiple inheritance.
- Isa triangles indicate the subclass relationship.
  - Point to the superclass.

  Note: A class/subclass relationship is often called an IS-A (or IS-AN) relationship

Subclass vs. superclass



Example: Subclasses



subclasses and specialization.

Three specializations of EMPLOYEE:
{SECRETARY, TECHNICIAN, ENGINEER}
{MANAGER}
{HOURLY_EMPLOYEE, SALARIED_EMPLOYEE}

Specialization vs. Generalization

- Specialization
  - Define a set of subclasses of an entity type
  - Establish additional specific attributes with each subclass
  - Establish additional specific relationship types between each subclass and other entity types or other subclasses

Specialization vs. Generalization

Eg: the set of subclasses {SECRETARY, ENGINEER, TECHNICIAN} is a specialization of the superclass EMPLOYEE that distinguishes among employee entities based on the job type of each employee entity

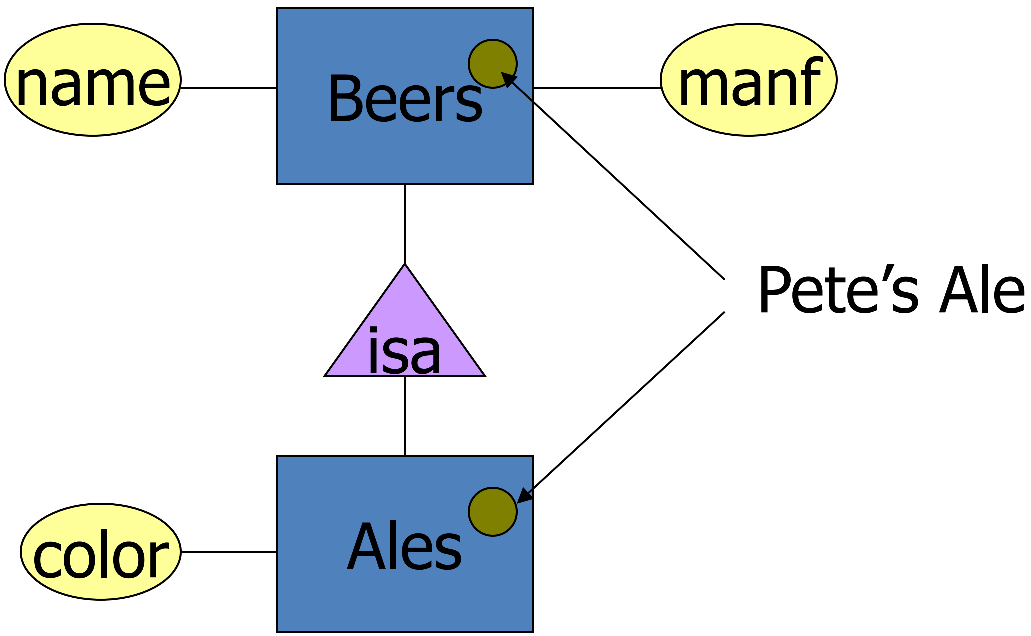
Specialization vs. Generalization

- Generalization: reverse process of abstraction in which we suppress the differences among several entity types, identify their common features, and generalize them into a single superclass of which the original entity types are special subclasses

E/R Vs. Object-Oriented Subclasses

- In OO, objects are in one class only.
  - Subclasses inherit from superclasses.
- In contrast, E/R entities have *representatives* in all subclasses to which they belong.
  - Rule: if entity *e* is represented in a subclass, then *e* is represented in the superclass (and recursively up the tree).

Example: Representatives of Entities



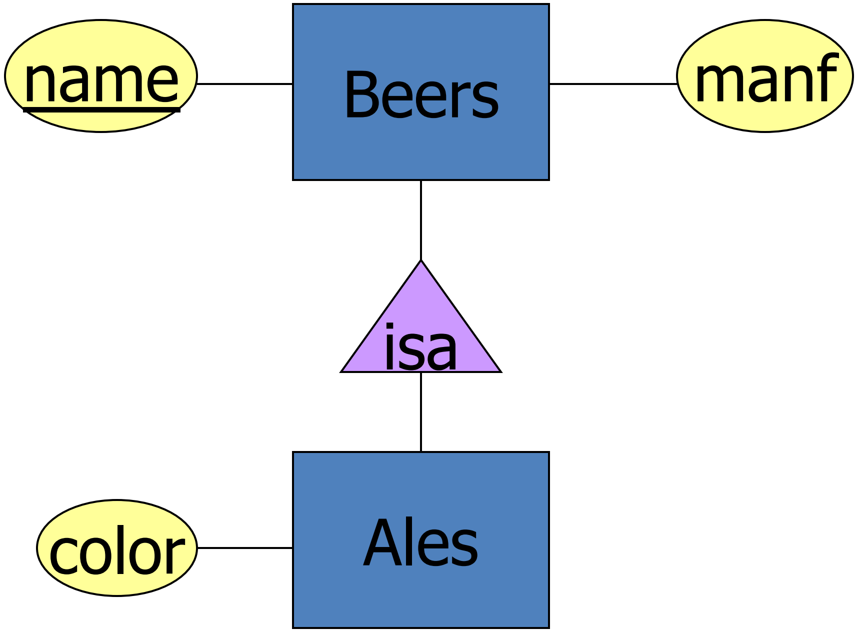
Keys

- A *key* is a set of attributes for one entity set such that no two entities in this set agree on all the attributes of the key.
  - It is allowed for two entities to agree on some, but not all, of the key attributes.
- We must designate a key for every entity set.

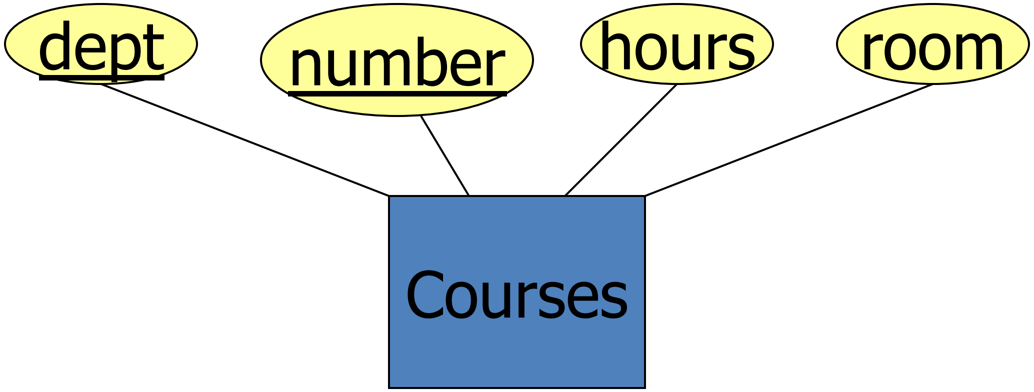
Keys in E/R Diagrams

- Underline the key attribute(s).
- In an Isa hierarchy, only the root entity set has a key, and it must serve as the key for all entities in the hierarchy.

Example: name is Key for Beers



Example: a Multi-attribute Key



- Note that hours and room could also serve as a key, but we must select only one key.

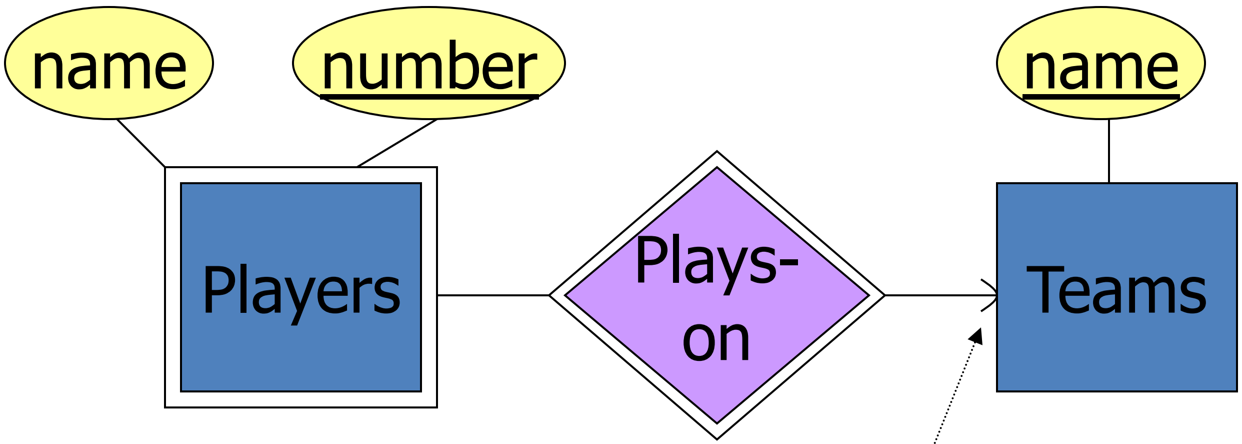
Weak Entity Sets

- Occasionally, entities of an entity set need “help” to identify them uniquely.
- Entity set *E* is said to be *weak* if in order to identify entities of *E* uniquely, we need to follow one or more many-one relationships from *E* and include the key of the related entities from the connected entity sets.

Example: Weak Entity Set

- name is almost a key for football players, but there might be two with the same name.
- number is certainly not a key, since players on two teams could have the same number.
- But number, together with the team name related to the player by Plays-on should be unique.

In E/R Diagrams



Note: must be rounded because each player needs a team to help with the key.

- Double diamond for *supporting* many-one relationship.
- Double rectangle for the weak entity set.

Weak Entity-Set Rules

- A weak entity set has one or more many-one relationships to other (supporting) entity sets.
  - Not every many-one relationship from a weak entity set need be supporting.
  - But supporting relationships must have a rounded arrow (entity at the "one" end is guaranteed).

Weak Entity-Set Rules – (2)

- The key for a weak entity set is its own underlined attributes and the keys for the supporting entity sets.
  - E.g., (player) number and (team) name is a key for Players in the previous example.

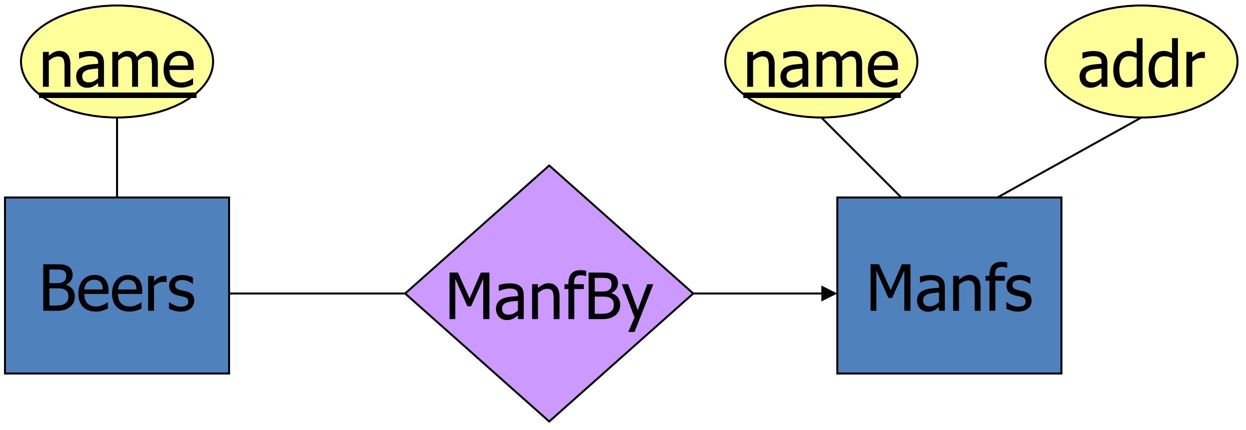
Design Techniques

1. Avoid redundancy.
2. Limit the use of weak entity sets.
3. Don’t use an entity set when an attribute will do.

Avoiding Redundancy

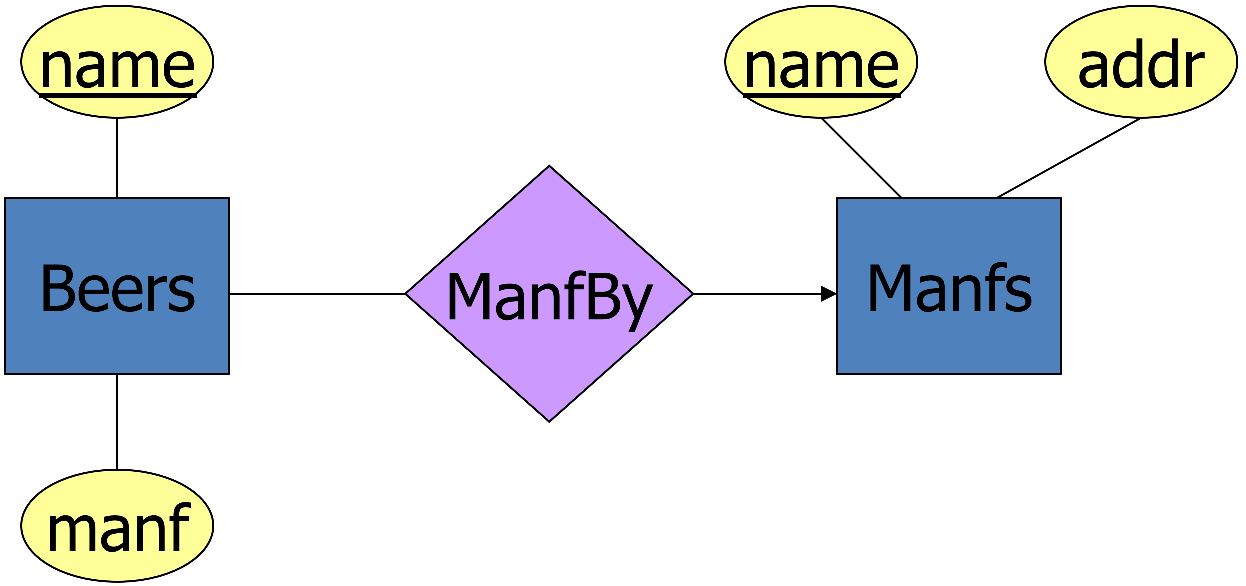
- *Redundancy* = saying the same thing in two (or more) different ways.
- Wastes space and (more importantly) encourages inconsistency.
  - Two representations of the same fact become inconsistent if we change one and forget to change the other.
  - Recall anomalies due to FD's.

Example: Good



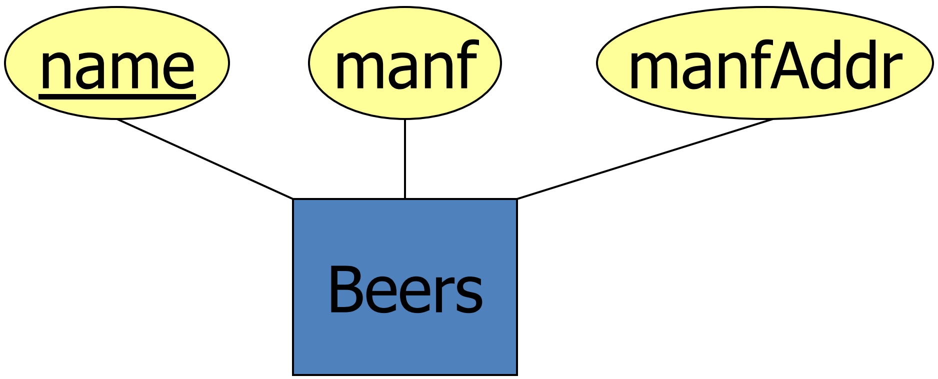
This design gives the address of each manufacturer exactly once.

Example: Bad



This design states the manufacturer of a beer twice: as an attribute and as a related entity.

Example: Bad



This design repeats the manufacturer's address once for each beer and loses the address if there are temporarily no beers for a manufacturer.

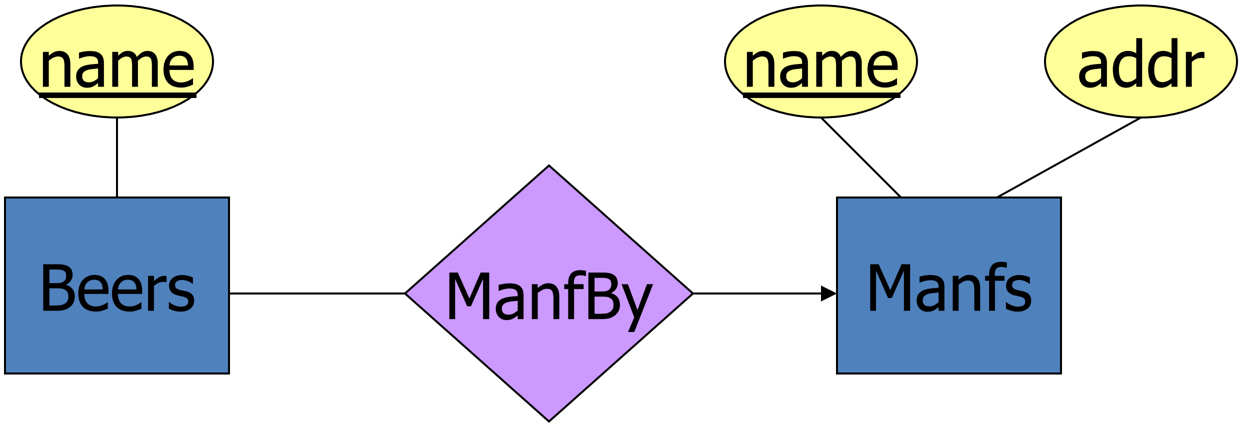
Entity Sets Versus Attributes

- An entity set should satisfy at least one of the following conditions:
  - It is more than the name of something; it has at least one nonkey attribute.

    or

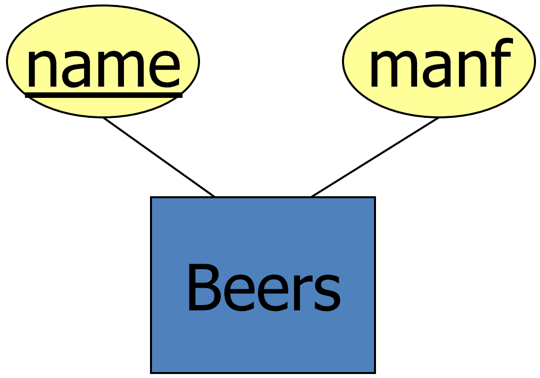
  - It is the “many” in a many-one or many-many relationship.

Example: Good



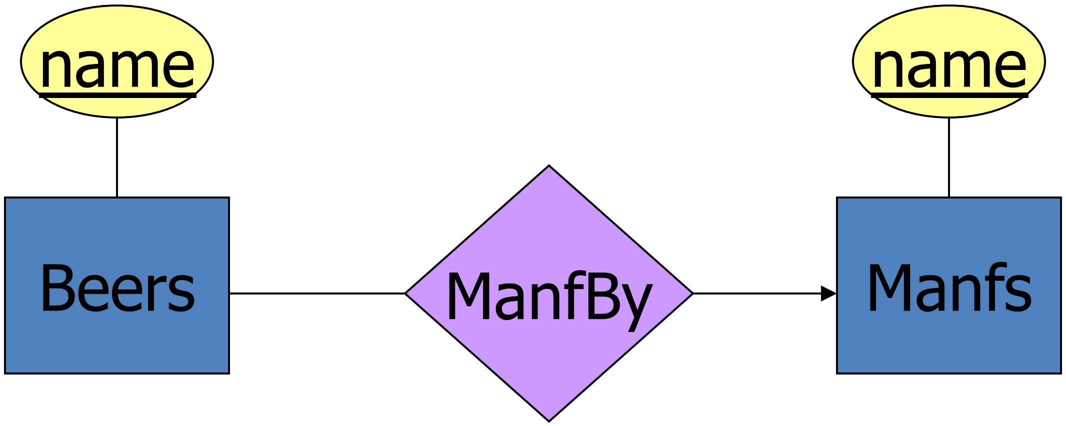
- Manfs deserves to be an entity set because of the nonkey attribute addr.
- Beers deserves to be an entity set because it is the "many" of the many-one relationship ManfBy.

Example: Good



There is no need to make the manufacturer an entity set, because we record nothing about manufacturers besides their name.

Example: Bad



Since the manufacturer is nothing but a name, and is not at the "many" end of any relationship, it should not be an entity set.

Don’t Overuse Weak Entity Sets

- Beginning database designers often doubt that anything could be a key by itself.
  - They make all entity sets weak, supported by all other entity sets to which they are linked.
- In reality, we usually create unique ID’s for entity sets.
  - Examples include social-security numbers, automobile VIN’s etc.

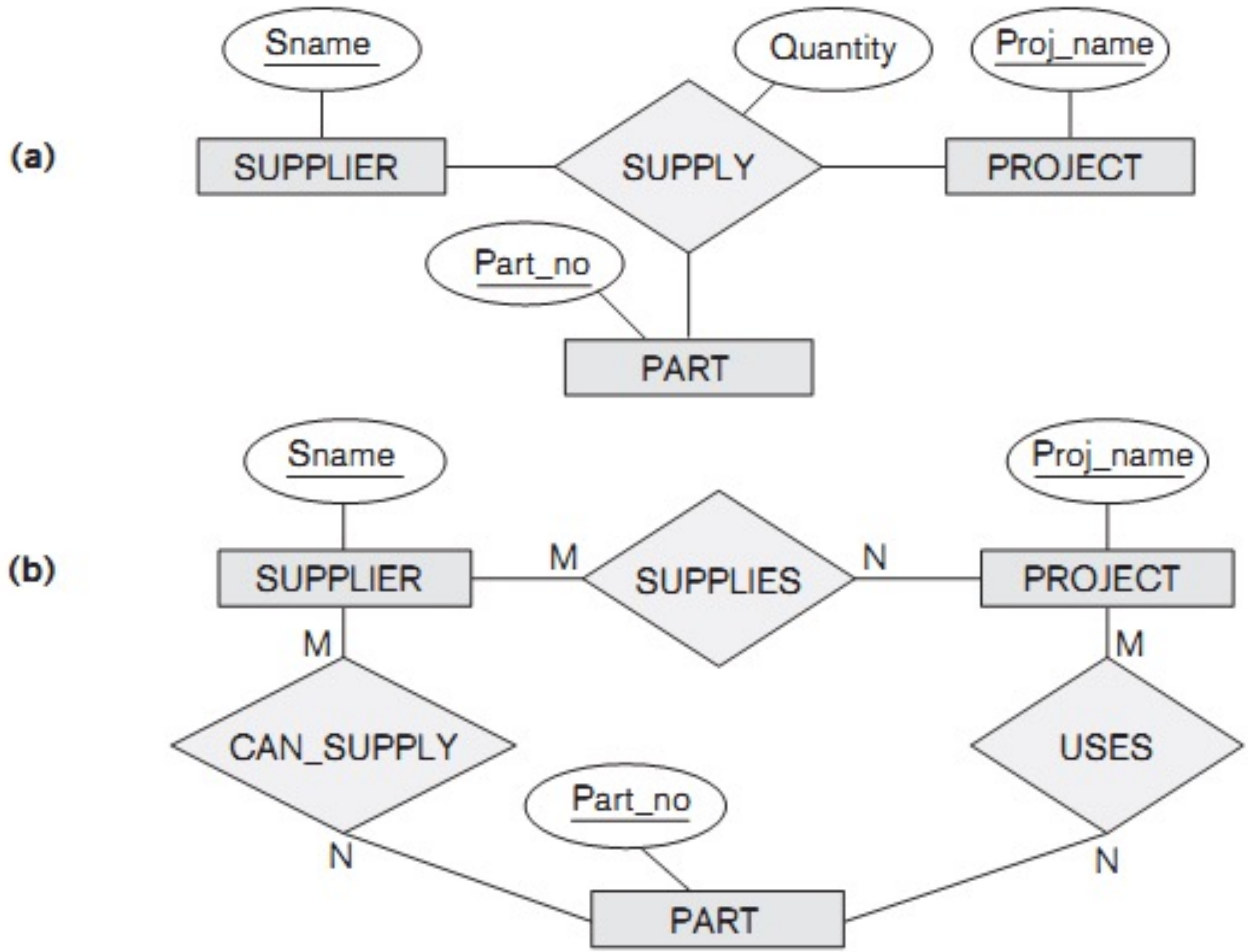
When Do We Need Weak Entity Sets?

- The usual reason is that there is no global authority capable of creating unique ID's.
- Example: it is unlikely that there could be an agreement to assign unique player numbers across all football teams in the world.

Degree of relationship

- Number of entities in a relationship
  - Unary
  - Binary
  - Ternary
  - n-ary

Degree of relationship



From E/R Diagrams to Relations

Step 1:

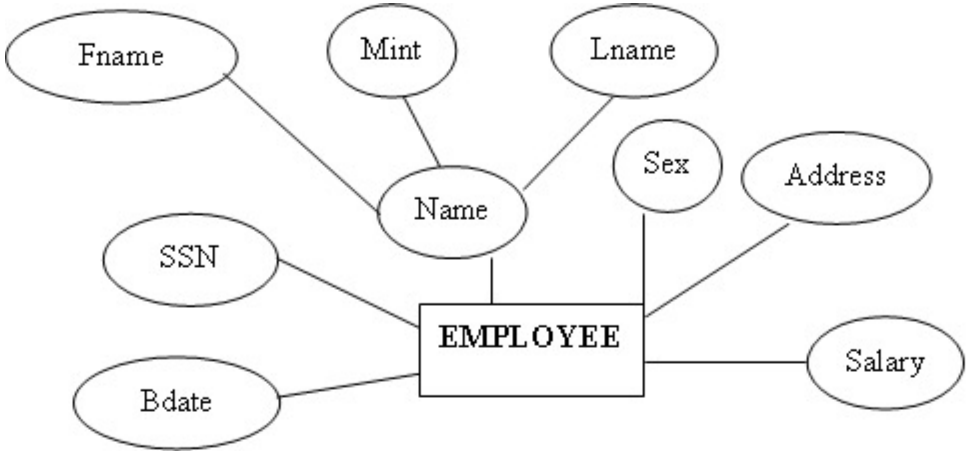
- Entity set -> relation.
  - Attributes -> attributes.
- Relationships -> relations whose attributes are only:
  - The keys of the connected entity sets.
  - Attributes of the relationship itself.

From E/R Diagrams to Relations

- Example

Entities: EMPLOYEE, DEPARTMENT, PROJECT

==> relations: EMPLOYEE, DEPARTMENT, PROJECT

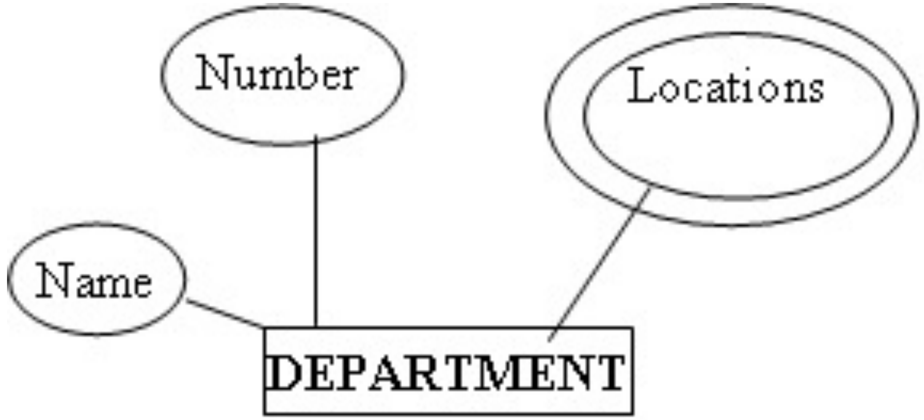


→ **Relation: EMPLOYEE (Ssn, fname, mint, lname, bdate, sex, address, salary)**

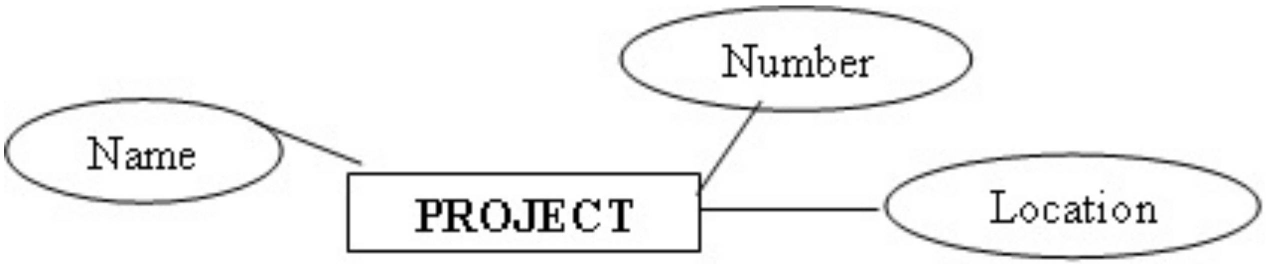
From E/R Diagrams to Relations

→ **Relation: DEPARTMENT (Dnumber, Dname)**

"Locations" is a multi-values attribute, then it will become a relation.



→ **PROJECT (Pnumber, Pname, Plocation)**

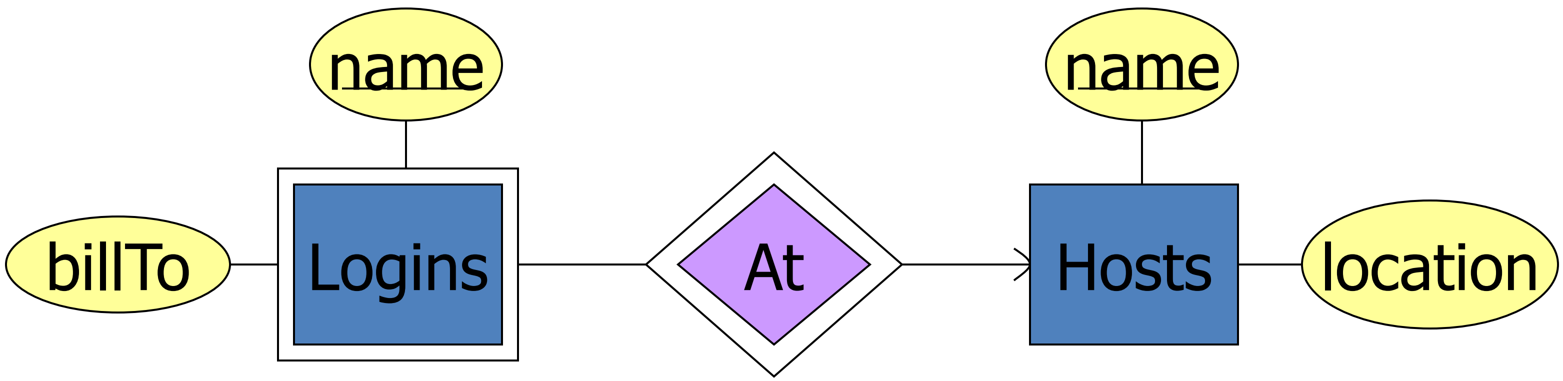


From E/R Diagrams to Relations

Step 2: handling weak entity

- For each Weak Entity in ERD:
  - Create a relation R for weak entity
    - Attributes → attributes
    - identity attribute of supporting entity → foreign key of R
    - Primary key of R is the combination of the identity attributes of weak entity and supporting entity

Example: Weak Entity Set -> Relation



Hosts(hostName, location)

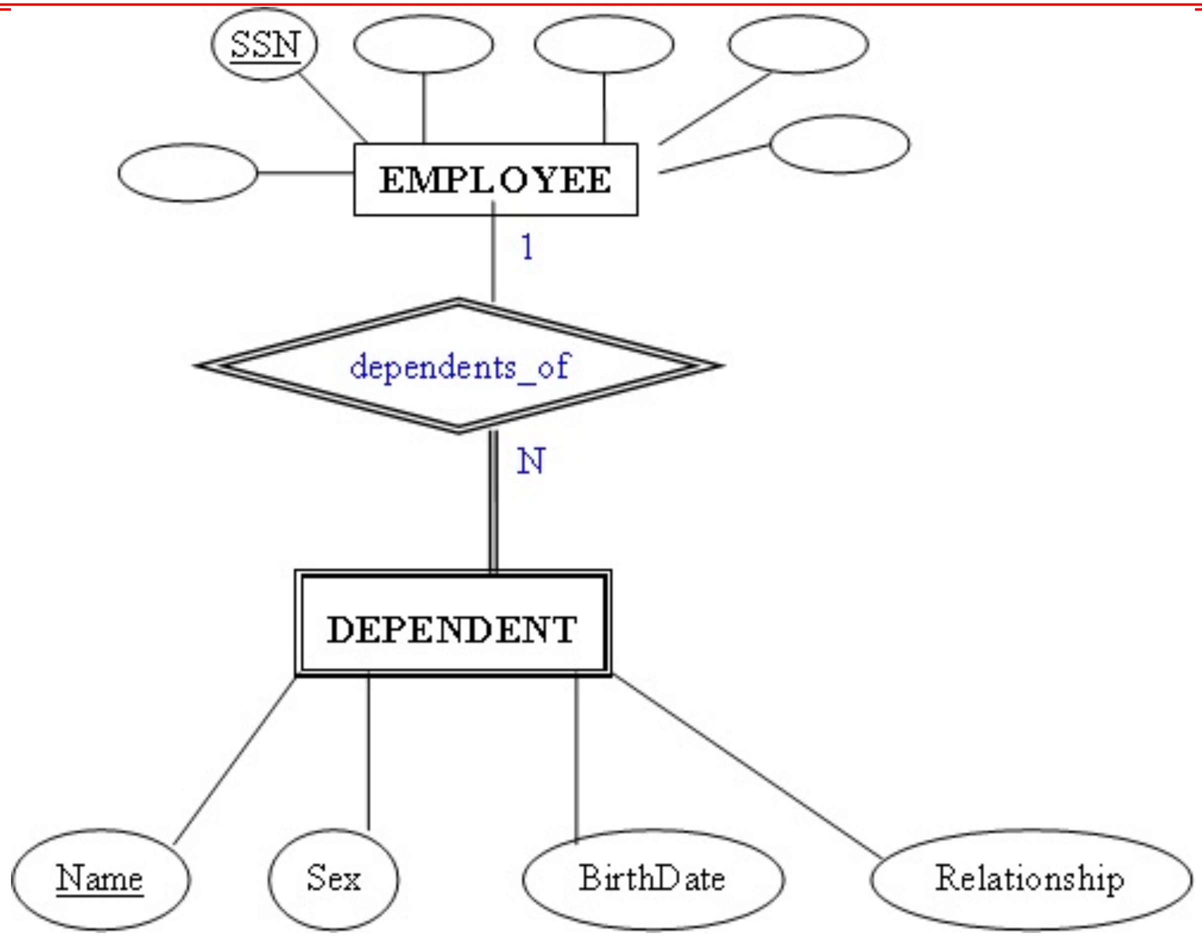
Logins(loginName, hostName, billTo)

~~At(loginName, hostName, hostName2)~~

At becomes part of Logins

Must be the same

From E/R Diagrams to Relations



→ **DEPENDENT (Essn, Dependent_name , sex, bdate, relationship)**

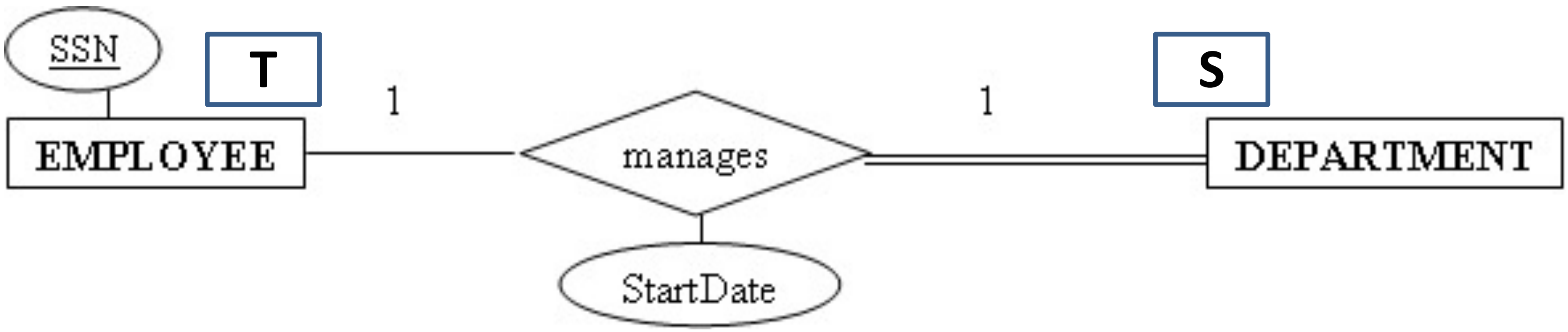
From E/R Diagrams to Relations

Step 3

For each one-to-one relationship in ERD:

- Create a relation S_T. The entity with mandatory participant becomes S, another one is T.
- Take primary key of T to become foreign key of S.
- Attributes of this relationship becomes attributes of S.

From E/R Diagrams to Relations

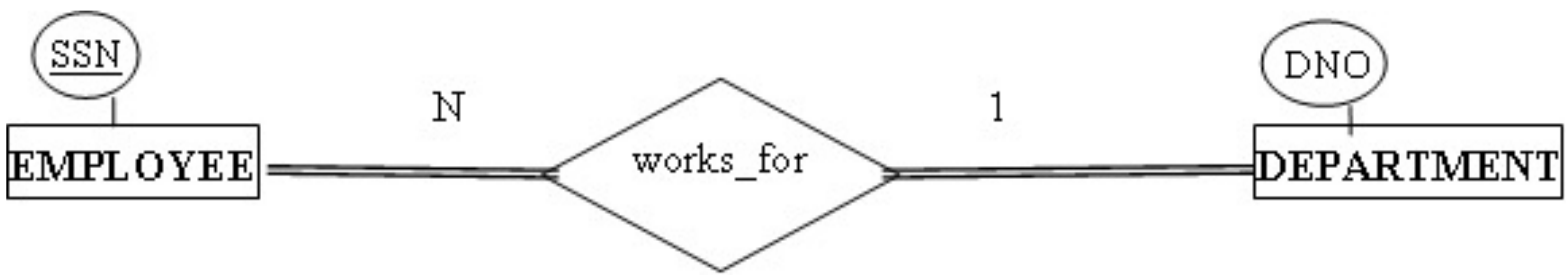


→ **DEPARTMENT(..., <u>MSSN</u>, MSTARTDATE)**

From E/R Diagrams to Relations

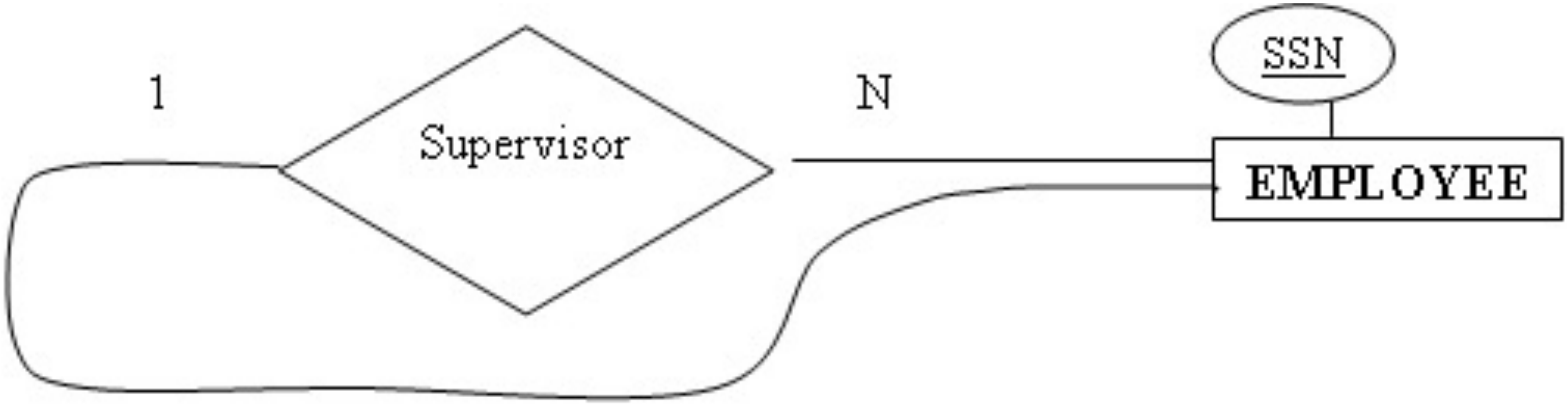
Step 4

- For each one-to-many(1:n) relationship in ERD:
  - Take primary key of 1-relation to be foreign key of n-relation



→ **EMPLOYEE(..., <u>DNO</u>)**

From E/R Diagrams to Relations



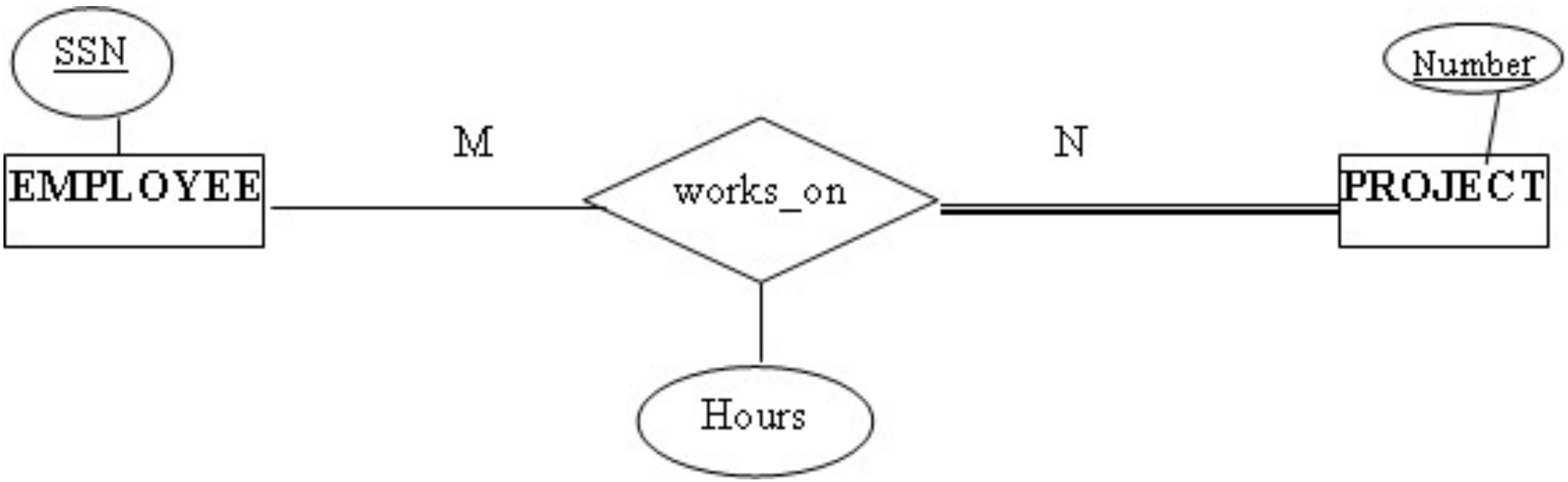
→ **EMPLOYEE(..., <u>SuperSSN</u>)**

From E/R Diagrams to Relations

Step 5

- For each many-to-many(M:N) relationship
  - Create a new relation R,
  - Primary keys of two relations in this relationship are foreign keys of R.
  - Primary key of are is the combination of these foreign keys.

From E/R Diagrams to Relations



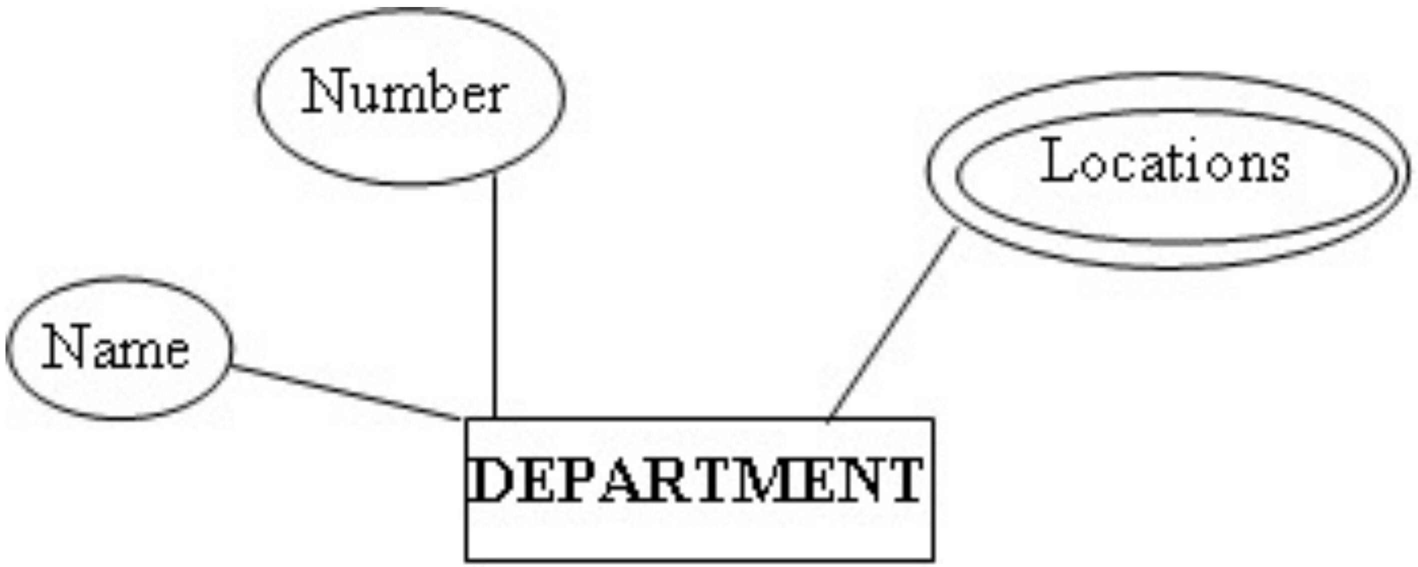
WORKS_ON(<u>PNO, ESSN</u>, Hours)

From E/R Diagrams to Relations

Step 6

- For multi-value attribute:
  - This attribute becomes a relation R.
  - Identity attribute of the main entity becomes foreign key of R.
  - Primary key of R is the combination of its primary key and the above foreign key.

From E/R Diagrams to Relations



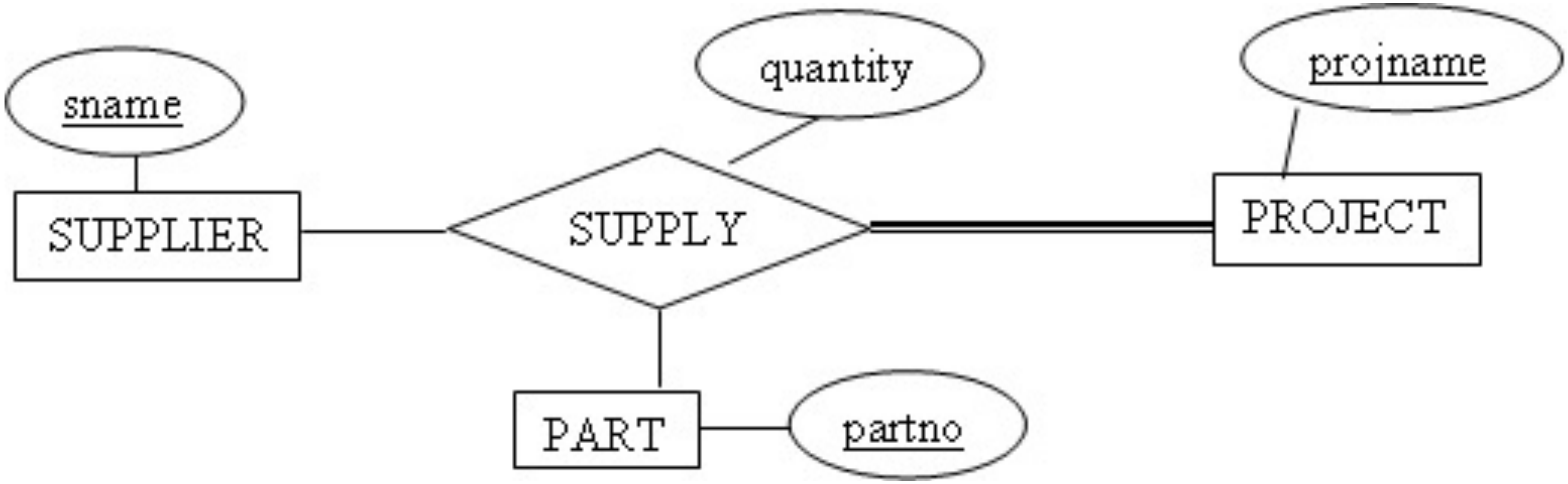
→ **DEPT_LOCATIONS (<u>DNumber, DLocation</u>)**

From E/R Diagrams to Relations

Step 7

- Relationship type of degree higher than 2
  - Create a new relation R,
  - Primary keys of all relations in the relationship are foreign keys of R.
  - These foreign keys are primary key of R.

From E/R Diagrams to Relations



→ **SUPPLY (<u>SName, ProjName, PartNo</u>, Quantity)**

From E/R Diagrams to Relations

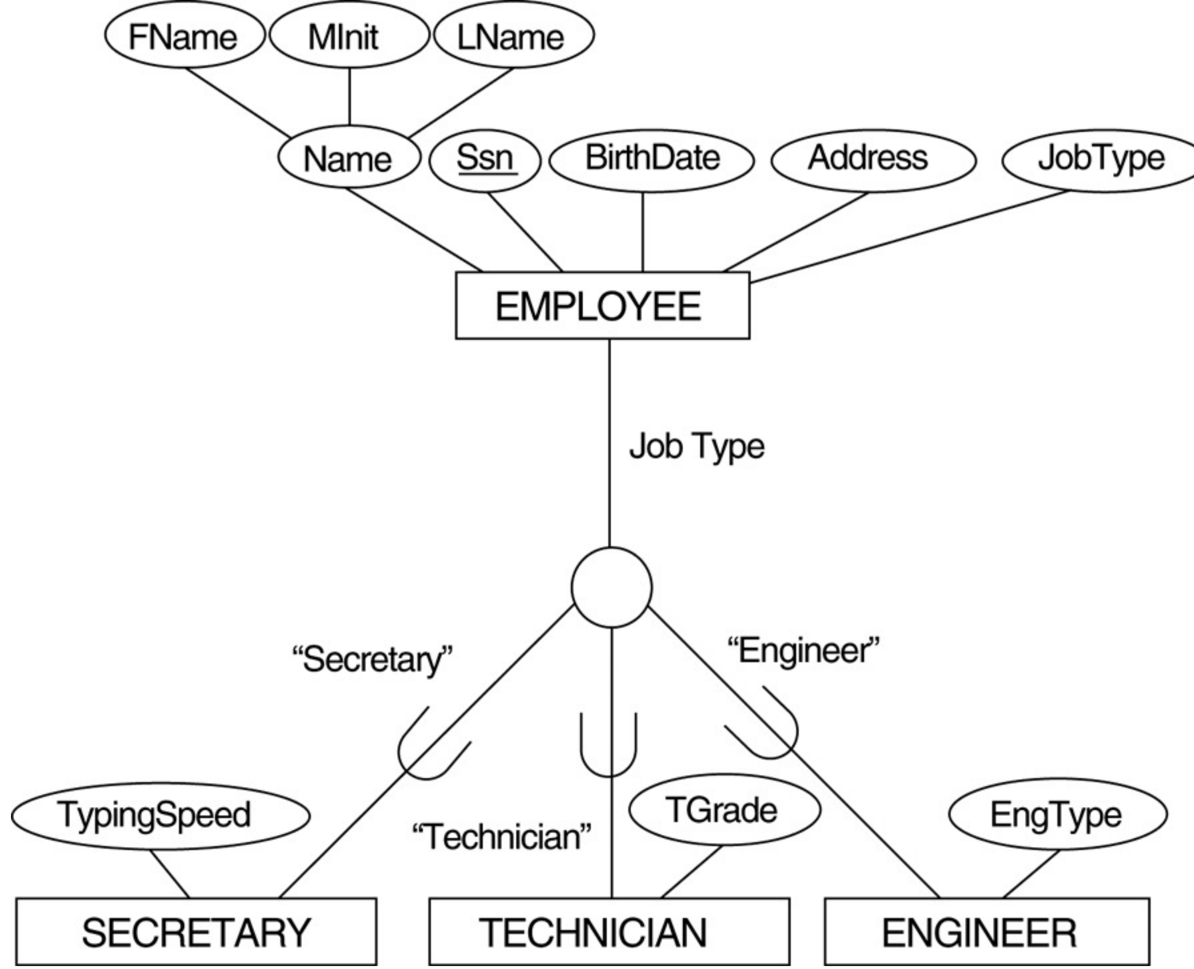
- Step8: Subclass
  - Superclass C with attributes {**k**, a_1, a_2…, a_n}
  - Subclass {S_1, S_2,..,S_m}

Step 8 – option 1

- Superclass C with attributes {**<u>k</u>**, a_1, a_2…, a_n}
- Subclass {S_1, S_2,..,S_m}

- Create relation L for superclass C with attributes: **Attrs(L)={k, a_1, …, a_n}** and primary key of L is: **PK(L)=k.**
- Create relation L_i for each subclass S_i with **Attrs(L_i)={k} U {attributes of S_i}** and primary key of L_i is **PK(L_i)=k.**

→ specialization:



(a) EMPLOYEE

SSN	FName	MInit	LName	BirthDate	Address	JobType

SECRETARY

SSN	TypingSpeed

TECHNICIAN

SSN	TGrade

ENGINEER

SSN	EngType

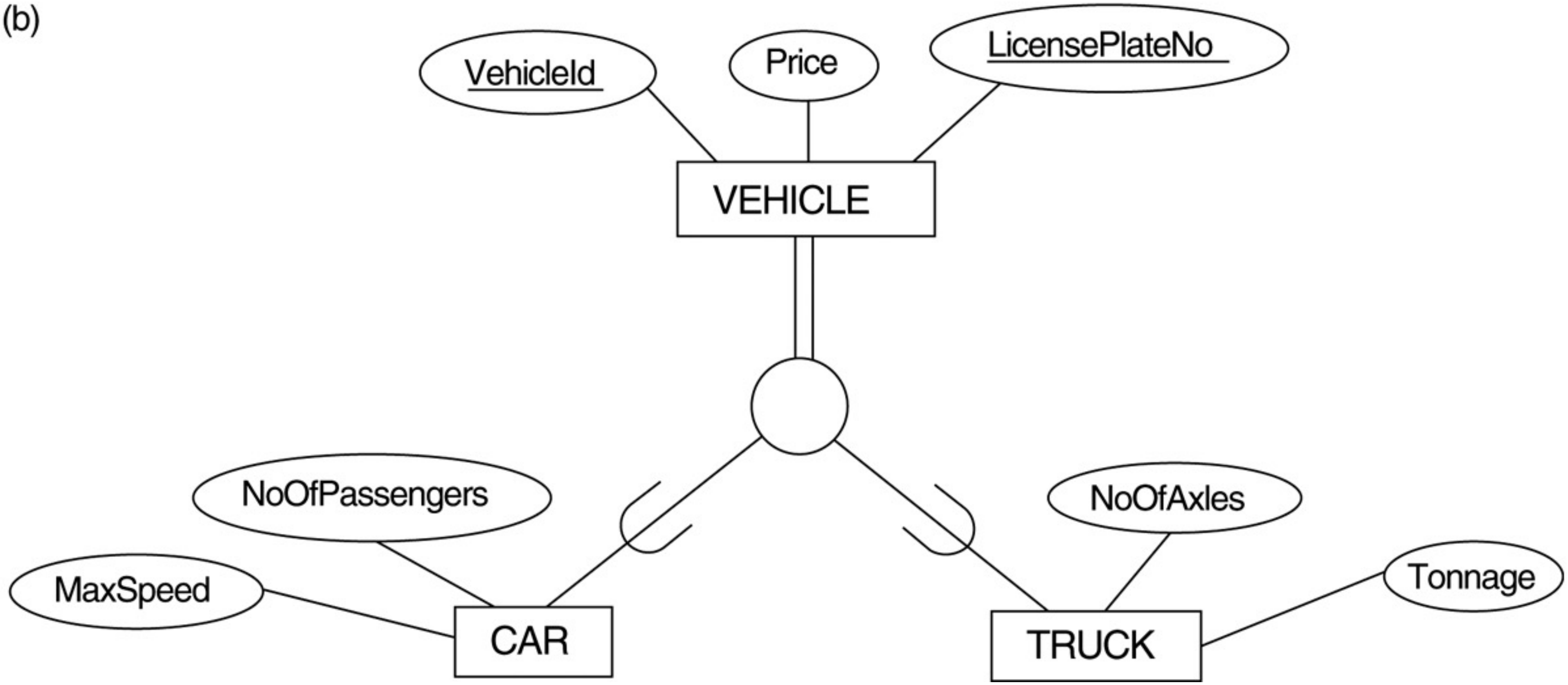
Step 8 – option 2

- Superclass C with attributes {**<u>k</u>**, a_1, a_2…, a_n}
- Subclass {S_1, S_2,..,S_m}

• Create relations L_i for each subclass S_i, with **Attr(L_i) = {k, a_1, a_2,…, a_m} U {attributes of S_i} và PK(L_i) = k.**

Generalization. (b) Generalizing CAR and TRUCK into the superclass VEHICLE.

(b)



(b) CAR

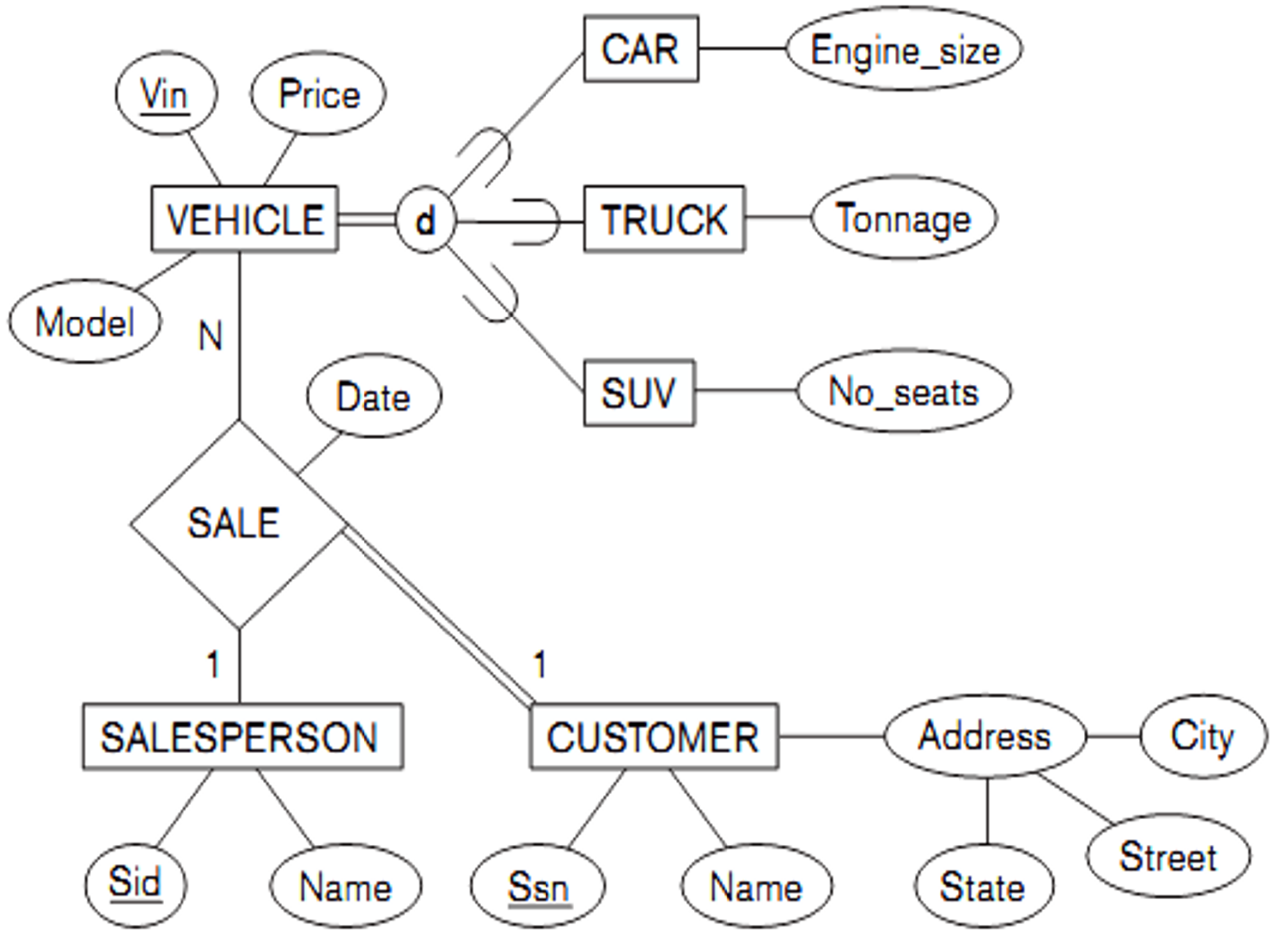
VehicleId	LicensePlateNo	Price	MaxSpeed	NoOfPassengers

TRUCK

VehicleId	LicensePlateNo	Price	NoOfAxles	Tonnage

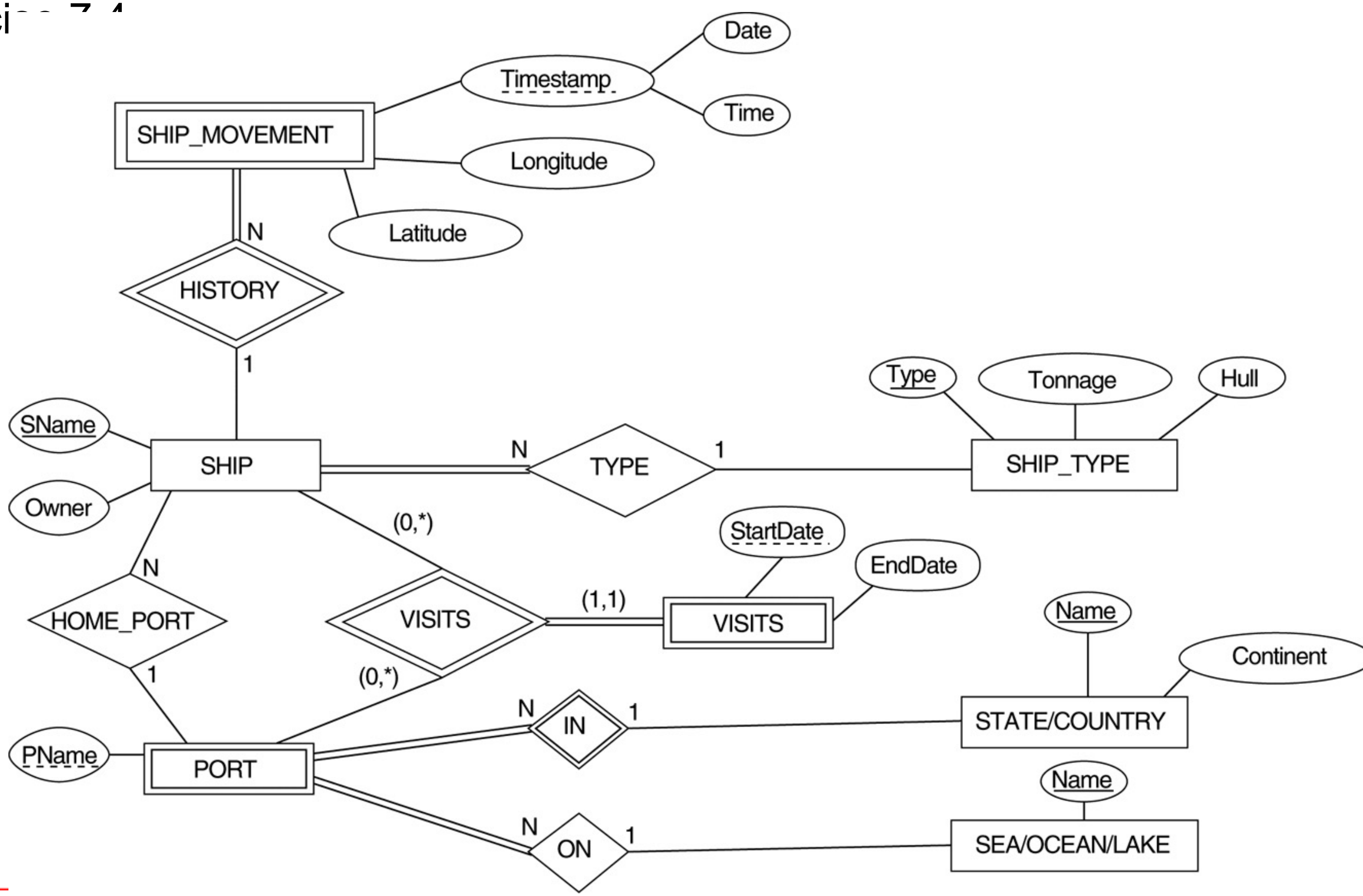
Exercises

Mapping ERD to relations

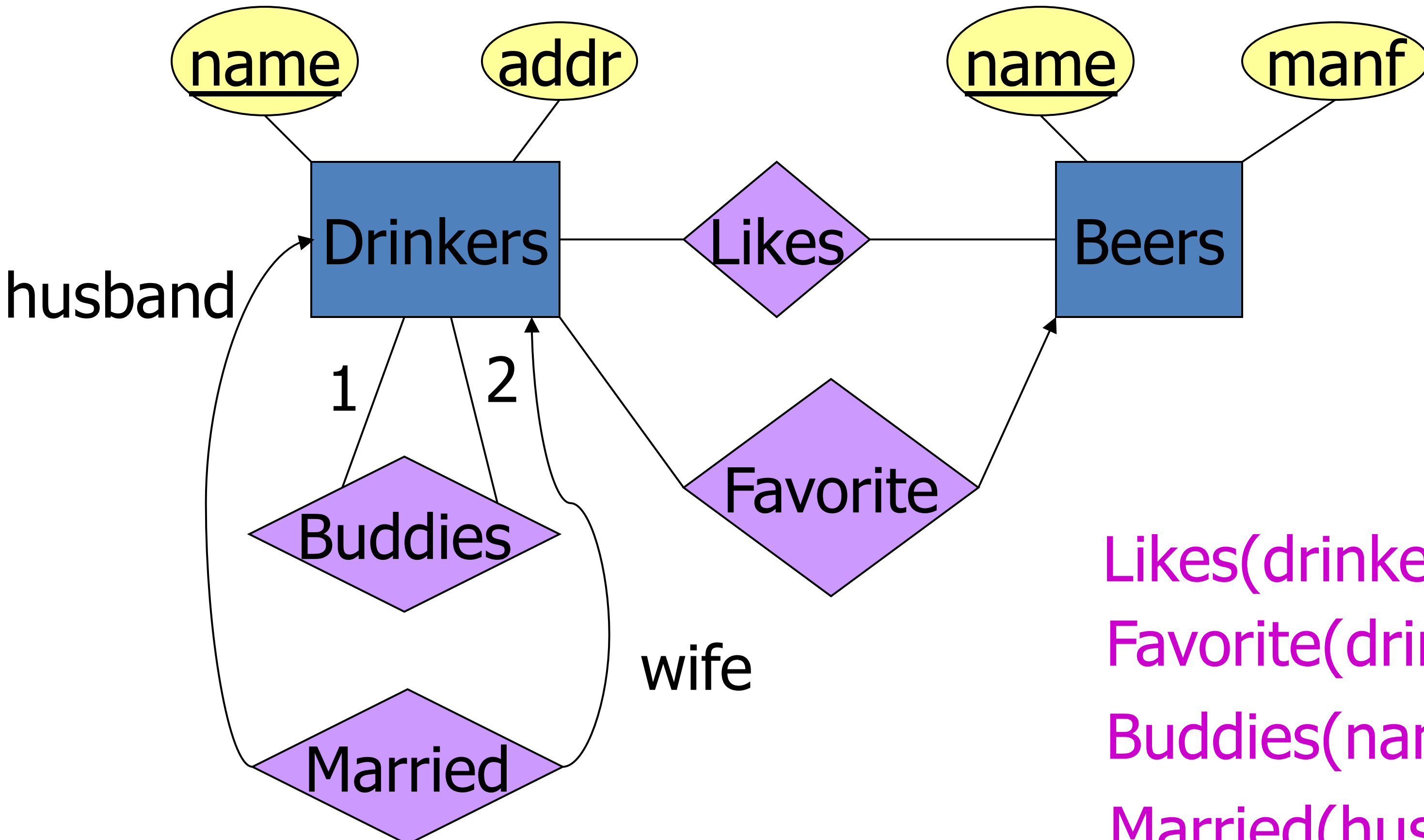


Mapping Exercise

Exercise 7.4



Relationship -> Relation



Likes(drinker, beer)
Favorite(drinker, beer)
Buddies(name1, name2)
Married(husband, wife)

Combining Relations

- OK to combine into one relation:
  1. The relation for an entity-set *E*
  2. The relations for many-one relationships of which *E* is the "many."
- Example: Drinkers(name, addr) and Favorite(drinker, beer) combine to make Drinker1(name, addr, favBeer).

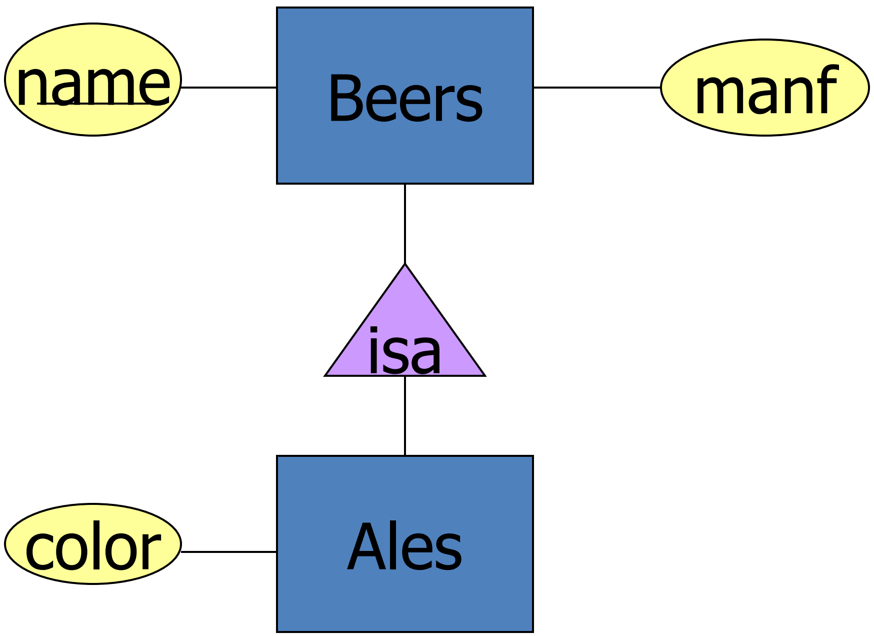
Risk with Many-Many Relationships

- Combining Drinkers with Likes would be a mistake. It leads to redundancy, as:

name	addr	beer
Sally	123 Maple	Bud
Sally	123 Maple	Miller

Redundancy

Example: Subclass -> Relations



Object-Oriented

name	manf
Bud	Anheuser-Busch

Beers

name	manf	color
Summerbrew	Pete’s	dark

Ales

Good for queries like “find the color of ales made by Pete’s.”

E/R Style

name	manf
Bud	Anheuser-Busch
Summerbrew	Pete’s

Beers

name	color
Summerbrew	dark

Ales

Good for queries like “find all beers (including ales) made by Pete’s.”

Using Nulls

name	manf	color
Bud	Anheuser-Busch	NULL
Summerbrew	Pete’s	dark

Beers

Saves space unless there are *lots* of attributes that are usually NULL.